Transportation Research Part A 208 (2026) 104962

Contents lists available at ScienceDirect

Transportation Research Part A

journal homepage: www.elsevier.com/locate/tra

# Desirable bikeshare routes: Nonlinear impacts of micro-level street environments

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ARTICLE INFO

*Keywords:*
Dockless bikeshare (DBS)
Cyclists
Route choice
Non-linear effects
Micro-level street environment
Road damage detection

ABSTRACT

Facilitating an attractive and safe street environment (SE) for cyclists can yield multifaceted benefits for sustainable, healthy cities. However, due to limited urban-scale trajectory data, the influence of micro-level SE (e.g., trees, shrubs, perceptions, and road cracks) on cyclists' route preferences remains less understood. Using over 4,000 Lime Dockless Bikeshare trajectories, this study constructed a Desirable Bikeshare Index (DBI) that measures differences between observed trips and the shortest routes suggested by Google Maps at the street-segment level. Using computer vision, road damage was detected in street view images, alongside objective streetscape features and subjective perceptions (e.g., enclosure) to characterize micro-level SEs comprehensively. Nonlinear associations between SE and DBI were identified using explainable machine learning algorithms, and customer satisfaction theory (e.g., basic, excitement, performance factors) was adopted to interpret the threshold effects. Contrary to earlier findings, green spaces do not always attract cyclists – lush trees draw cyclists, while too many shrubs repel them. Moreover, sidewalks are a must-have (basic factor), while road damage is more common where cyclists aggregate, underscoring the need for timely monitoring and maintenance of bicycle infrastructure. Perceptions of accessibility, order, and richness are excitement factors that explain route desirability. Interestingly, perceived ecology and enclosure promote cycling at low values, then quickly play a negative role beyond mid-to-high values. Our findings underscore the need to combine subjective perceptions with objective streetscapes to delineate a comprehensive micro-level SE. Urban designers and transport planners can leverage revealed thresholds to implement cost-effective infrastructure upgrades that facilitate desirable street environments for cyclists.

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https://doi.org/10.1016/j.tra.2026.104962
Received 18 September 2024; Received in revised form 20 February 2026; Accepted 10 March 2026

## 1. Introduction

### 1.1. Route choice studies enabled by GPS data

As an emerging active travel mode, bikeshare, especially dockless bike-sharing (DBS), provides flexible, affordable, and sustainable alternatives to conventional mobility options (Shaheen et al., 2021). The prevalence of DBS further prompted the government to foster a cycling-friendly environment through urban design policies and interventions (Pucher and Buehler, 2017). For instance, during the pandemic, Toronto installed pop-up cycle tracks to facilitate cycling as an alternative to public transit for commuting, many of which were later made permanent (Mandhan and Gregg, 2023). Planning and transportation scholars have increasingly focused on the built environment (BE) attributes that enhance street bikeability to better understand what dimensions warrant investment to promote active mobility. These BE attributes influence cyclists' travel behaviors, including cycling propensity (Wang et al., 2020), trip volume (Gao and Fang, 2025; Wei et al., 2023; Zhang et al., 2025), and route choice (Fitch and Handy, 2020; Menghini et al., 2010; Scott et al., 2021).

Route choice research offers planners and policymakers valuable insights by revealing the underlying trade-offs cyclists face when making discrete decisions (Zimmermann et al., 2017). Two mainstream methods have relied on either collecting stated preferences to map out route choices visually (Vedel et al., 2017; Winters et al., 2010) or revealed preferences through field observations or passive trajectory recordings using distributed GPS devices (Broach et al., 2012). The latter approach has recently received renewed interest due to the integration of GPS into bikeshare systems (Chung et al., 2024; Cubells et al., 2023; Khatri et al., 2016). In theory, researchers assume that people choose the shortest routes to minimize the travel time cost to maximize the utility (Meyer, 2016), while extensive studies have indicated that cyclists do not necessarily opt for the shortest path (Broach et al., 2012; Fitch and Handy, 2020; Zimmermann et al., 2017). However, when cyclists detour from the shortest paths, their choices may reflect local responses to the BE or other factors, suggesting poorly understood incentives that drive such deviations that can compensate for the additional time costs (Oyama, 2024; Park and Akar, 2019). Understanding specific environmental characteristics that affect bikeshare users' route choice preferences to deviate from the theoretically shortest routes can offer direct evidence to inform practical urban design strategies for facilitating a bicycle-friendly street environment that encourages active travel towards healthy and human-centric planning (Cubells et al., 2023; Sevtsuk et al., 2021; Zhao et al., 2024).

Despite these advancements, existing studies have three key limitations that remain to be addressed: (1) insufficient understanding of DBS users' detour behaviors and their explanatory factors, (2) predominant focus on macro-level BE while overlooking micro-level street characteristics, and (3) limited understanding of nonlinear effects of BE on route preferences. These research gaps motivate our study to comprehensively investigate how micro-level street environment (SE) attributes shape DBS route choices through interpretable machine learning approaches.

### 1.2. Research objective

Our main objective is to elucidate the understudied nonlinear impact of micro-level SE on DBS users' deviation from the shortest path, providing insights for planning and designing bicycle-friendly streets and enriching our understanding beyond the effects of macro-level BE. The contributions are threefold: 1) Inspired by prior literature (Salazar Miranda et al., 2021), we developed a Desirable Bikeshare Index (DBI) to assess the disparity between observed and simulated shortest routes based on over four thousand DBS trajectories. Street segments receive a positive increment when traversed by actual trips and a negative increment when appearing exclusively in simulated shortest paths. Therefore, aggregated scores capture revealed detours from shortest-path routing as a proxy for route desirability, though they may understate preferences for segments that frequently overlap with shortest routes. 2) This study comprehensively investigated how micro-level SE attributes may complement conventional macro-level BE factors in affecting route deviation preferences. In addition to the neglected subjective perceptions and objective street features, we also trained a deep learning (DL) model to effectively detect road surface maintenance conditions. 3) Leveraging explainable Machine Learning (ML) algorithms, we applied the three-factor analysis (basic/excitement/performance) from customer satisfaction studies to interpret the nonlinear associations between micro-level SE characteristics and detour desirability. This approach offers new perspectives on the complex decision-making processes of DBS users in smaller cities. The empirical evidence informs the potential for more human-centric routing recommendation systems, for upgrading bike-share facility planning, and for designing cycling-friendly streets in pursuit of sustainability goals.

To systematically address these objectives, the remainder of this paper is organized as follows. Section 2 reviews the existing literature on route choice behaviors, the influences of the built environment, and nonlinear effects method approaches, concluding with a systematic synthesis of research gaps. Section 3 describes our research design, including the analytical framework, study area, data sources, and methods for constructing key variables. Section 4 presents the empirical results from both regression and machine learning models. Section 5 discusses the theoretical and practical implications of our findings, along with their limitations and future research directions. Section 6 concludes with policy recommendations for promoting cycling-friendly urban design.

## 2. Literature review

### 2.1. Route choices and sustainable bikeshare

Bikeshare fosters equitable access to safe and reliable non-motorized transportation options, supporting sustainable mobility as a solution to last-mile travel for short-distance trips (Eren and Uz, 2020; Wang et al., 2024). As cities shift towards micro-mobility bikeshare systems, placing network infrastructure at key urban nodes and routes where cyclists prefer can significantly improve accessibility prediction to mode choice (Zimmermann et al., 2017) and promote cycling in general (Chou et al., 2023). Accordingly, strategically locating relevant bikeshare facilities in high-demand regions reduces reliance on motor vehicles and thereby significantly reduces greenhouse gas emissions and air pollution (Becker et al., 2022; Bullock et al., 2017; Chou et al., 2023). Therefore, gauging and interpreting model parameters that relate route choice to associated factors is crucial for understanding cyclists' trade-offs in situ (Zimmermann et al., 2017).

### 2.2. The influence of macro-level BE

Cycling route decision-making involves a complex interplay of psychological motives, habits, and environmental perceptions (Koh and Wong, 2013; Ma et al., 2014). For instance, the Random Utility Maximization model (Ben-Akiva and Lerman, 1985) has been pivotal in explaining cyclists' decisions when faced with multiple discrete choices, while Sener et al. (2009) identified BE attributes influencing bicycle route choices, such as cycling facilities (e.g., bike paths), and physical characteristics (e.g., slope).

Using emerging GPS data to reflect cyclists' revealed preferences, BE characteristics are increasingly acknowledged as crucial in shaping cyclists' route choices (Chou et al., 2023; Meister et al., 2023; Park and Akar, 2019). Infrastructure, including bicycle facilities, has been extensively studied and consistently shown to have a positive impact (Broach et al., 2012; Fitch and Handy, 2020; Menghini et al., 2010; Park and Akar, 2019). For instance, Meister et al. (2023) noted that bike lanes and dedicated paths are positively perceived by cyclists in influencing route choices, while Khatri et al. (2016) found that bikeshare users are prone to choose routes with signalized intersections. Similarly, studies revealed that physical conditions, including topography, are also significant determinants, whereas cyclists would avoid routes with steep slopes or roads with high traffic volumes due to safety concerns (Broach et al., 2012; Fitch and Handy, 2020; Park and Akar, 2019; Prato et al., 2018; Scott et al., 2021). Importantly, road network characteristics influence cyclists' detours from the shortest routes. Chou et al. (2023) reported that the detour ratio is positively associated with the straightness and centrality of road networks, echoing the conclusion of Lu et al. (2018).

Furthermore, the types of land use surrounding routes also play a crucial role. For instance, Park and Akar (2019) found that cyclists avoid passing through commercial areas due to potential mixed traffic conditions. Prato et al. (2018) found that different trip purposes lead to varied land-use preferences, with some individuals preferring to traverse residential areas. These findings are further supported by Fitch and Handy (2020), who noted that commercial lands may significantly influence cyclists' route decisions while acknowledging individual-level heterogeneity.

### 2.3. The impact of micro-level SE on route choices

However, our in-depth review of the route choice literature has revealed a significant research gap. The effects of micro-level SE characteristics remain poorly understood and under-addressed. In particular, there is a lack of studies examining essential perceptual dimensions, including SE objective physical features (e.g., street greenery, sky, road maintenance) and more importantly, SE subjective perceptions such as aesthetics and enclosure. These are crucial for understanding cyclists' route choices (Blitz, 2021; Guo and He, 2021).

This gap is partially due to the search for more feasible methods to quantify micro-level SE characteristics; using GIS or conventional surveys may result in a spatial mismatch in scale when coupled with highly granular street segment-level trajectory data. Luckily, the newly accessible SVI can be leveraged as a reliable source for auditing cycling routes, reflecting cyclists' visual ground truth (Vanwolleghem et al., 2014). Specifically, using DL algorithms, objective features such as trees and buildings can be semantically extracted, while subjective perceptions can be predicted by integrating SVI, Computer Vision (CV), and visual surveys (Dubey et al., 2016; Qiu et al., 2022; Ramírez et al., 2021).

Several recent studies have examined influences of objective features on various bikeshare behaviors, including trip volume, speed, and frequency (Wang et al., 2020; Wei et al., 2023; Zhang et al., 2023, 2025). For instance, Wang et al. (2020) set three buffer radii (500 m-1500 m) around subway stations in Shenzhen and revealed the positive impact of street greenery on DBS cycling frequency. However, the effect of these street characteristics on DBS route choices, specifically, decisions to deviate from the shortest path, remains under-investigated. The only relevant study by Zhao et al. (2024) constructed the preferred street index measure using actual DBS and shortest routes, and further found an insignificant relationship with street greenery but a positive correlation with streetlight facilities in Xiamen, China. However, the findings may not be broadly generalizable due to their focus on limited objective features and a short 4-hour time span. In this regard, several pedestrian route studies have examined how the perceived environment influences pedestrians' decisions to take detours from alternative routes (Basu and Sevtsuk, 2022; Salazar Miranda et al., 2021; Sevtsuk et al., 2021). For example, Sevtsuk et al. (2021) collected ~ 11,000 anonymous GPS-tracked walking routes in San Francisco and found that route choice is significantly associated with sky view but less with visual greenery. Salazar Miranda et al. (2021) introduced the Desirability index to measure pedestrians' willingness to deviate from the shortest path, suggesting that streets with sidewalks and enclosed street interfaces are more desirable.

Moreover, the potential impact of subjective perceptions (Song et al., 2025), such as enclosure, on route choices has yet to be explicitly addressed. While Ma et al. (2014) found that perceptions directly impact bicycling behaviors more than the objective environment, bikeability studies have started integrating subjective perceptions into their evaluation metrics (Ito and Biljecki, 2021). Nevertheless, the specific influence of these perceptions on route choice preferences remains unexplored. Investigating this aspect can provide unique empirical evidence that significantly enhances our understanding of perception studies (Guo and He, 2021; Ma and

Dill, 2016) and validate the concept of bikeability, potentially revolutionizing the field.

### 2.4. Examining nonlinear effects

Perceptual experiences can vary significantly with speed, influencing how environments are cognitively and affectively perceived (Nikiforiadis et al., 2021). For example, long-distance and high-speed cyclists might prefer a broader sky view and more trees, while wider streets with fewer visually disordered elements, such as large buildings and complex landscapes, are associated with higher riding volumes (Zhang et al., 2023). However, studies also reported that street greenery insignificantly correlates with route preferences (Zhao et al., 2024). These inconsistencies could stem from contextual differences or the complex nonlinearity between environmental characteristics and cycling behaviors. Notably, Zhang and Hu (2024) reported that DBS cycling distance positively correlates with perceived comfort, while its strength follows an upward trend. In contrast, perceived safety can exhibit a threshold effect, with its curve resembling a binary classifier. Similarly, Rui and Xu (2024) found that greenness initially promotes cycling but has a negative impact after a threshold is crossed. Such nuances may help corroborate discrepancies in previous research.

Appendix Table A1 lists key literature on route choices for active travel modes: walking, personal bikes, docked bikeshare, and DBS. It lacks empirical studies examining micro-level SE's influence on bicycling route choices, particularly for the DBS mode. Accordingly, the nonlinear impact of human-perceived objective features and subjective perceptions on DBS route choices and deviations from shortest paths should be systematically investigated.

### 2.5. Research gaps and contributions

Based on the comprehensive literature review above, this study aims to address three critical research gaps. First, the extent to which actual routes deviate from the shortest paths, and how much deviation preferences can be explained by BE, remains insufficiently understood in DBS mode. Although considerable research has examined detours from shortest or alternative routes using GPS trajectories from recruited cyclists (Broach et al., 2012; Meister et al., 2023; Park and Akar, 2019) or crowd-sourced data (Chou et al., 2023), these studies predominantly reflect the behaviors of bike owners, which may introduce bias when applied to bikeshare users' interactions with the BE. Moreover, even within bikeshare systems, DBS users display distinct routing behaviors and shorter excess time than station-based systems, which enforce more rigid rerouting mechanisms (Kou and Cai, 2021). Notably, emerging studies have examined the impact of BE factors (e.g., bike lanes) on route choices, though they have primarily focused on docked bikeshare systems (Chung et al., 2024; Cubells et al., 2023; Khatri et al., 2016). To our knowledge, the only DBS-related study is by Zhao et al. (2024), which developed the "preferred street index" to denote cycling route choice compared to the shortest path. Furthermore, most bikeshare route studies have been limited to short observation periods, typically lasting from several hours to a week (Chung et al., 2024; Zhao et al., 2024), underlining the need for more robust datasets for rigorous inference.

Second, existing studies have primarily focused on how macro-level BE factors such as land use types, infrastructure (e.g., bike lanes, traffic lights), and road conditions (e.g., topography) affect route choices (Cubells et al., 2023; Fitch and Handy, 2020; Menghini et al., 2010; Scott et al., 2021). Specifically, researchers reported that providing bike lanes could attract cyclists to detour from the shortest routes, while slopes may pose adverse impacts (Meister et al., 2023; Park and Akar, 2019). However, they largely overlook how bikeshare users interact with micro-level street environment characteristics, encompassing objective visual features and subjective perceptions. On the one hand, limited studies only revealed the positive influences of objective features, including streetlights and greenery extracted from street view images (SVI) (Zhao et al., 2024), warranting a more comprehensive investigation on how both ubiquitous (e.g., sky) and nuanced features (e.g., tree, bench, road surface maintenance) may influence cyclists' local decisions. Notably, while these objective features have been shown to impact pedestrian route choices (Salazar Miranda et al., 2021; Sevtsuk et al., 2021), their effects on cycling route decisions remain underexplored. On the other hand, while studies have examined how subjective perceptions of SE, including "enclosure" and "liveliness" affect aggregated cycling volume (Rui and Xu, 2024), they fail to control baseline volumes, and their influence on individual route choices remains unclear. We hypothesize that, in addition to conventional macro-level BE characteristics, these micro-level SE features could play a critical role in shaping DBS users' decisions to deviate from the shortest routes.

Lastly, research exploring the nonlinear effects of micro-level SE characteristics on cyclists' route choices remains sparse. Although a few studies have examined the nonlinear effects of subjective perceptions on influencing cycling volume (Rui and Xu, 2024) and distance (Zhang and Hu, 2024), such insights have yet to be extended to explore the nonlinear threshold effects on route preferences (Flanagan et al., 2016; Ma and Dill, 2016; Prato et al., 2018). Given the inherent complexity of how people perceive the SE and its interactions with other BE variables, nonlinear ML models (e.g., Random Forests) may better capture these variations in their effects, thereby facilitating a deeper understanding of the decision mechanism. In short, examining the subtle nonlinear impacts of micro-level SE attributes on route decisions is crucial to fully interpreting the nuanced interplay between the environment and human behavior.

## 3. Research design

### 3.1. Analytical framework

The analytical framework consists of six steps (Fig. 1). First, trajectories and corresponding shortest paths are aggregated at the street level. Second, recognizing that trip volume often fails to control for baseline effects and introduces bias due to mechanical factors, such as location effects in which cyclists may choose specific routes simply because of their location, we developed the

Desirable Bikeshare Index. The DBI reflects cyclists' deviations from the shortest path, capturing their revealed preferences and controlling for these mechanical effects. Third, macro-level BE variables, such as space syntax indicators and land uses, are collected to isolate their impacts. Sociodemographic factors are collected through census data. Fourth, SE attributes are evaluated using SVI and ML techniques, including analyses of objective streetscape elements extracted from CV algorithms and subjective perceptions derived from pairwise expert surveys. Additionally, road surface condition is evaluated using a Convolutional Neural Network (CNN). Fifth, we employ Ordinary Least Squares (OLS) regression models to compare the explanatory power of each variable group concerning the DBI. Finally, the study evaluates multiple ML models (e.g., XGBoost) and combines SHAP and Partial Dependence Plots (PDP) to interpret global feature importance and nonlinear threshold effects. This approach provides an unbiased and more flexible interpretation of the factors influencing deviations in DBS route choices.

### 3.2. Study area and trajectory data

The study area is Ithaca, a city in Upstate New York, which is a typical college town home to Cornell University. The actual trajectories used for this study were supplied by Lime, a bikeshare company (https://www.li.me/en-co), which partnered exclusively with the city to provide bikeshare services in 2018. Our trajectory data spans from November 21, 2019, to March 17, 2020, encompassing 4,430 valid trips (after data cleansing) and reflecting commuter behavior before the pandemic, based on secondary data from Qiu and Chang (2021). The trajectory preprocessing from raw GPS into remapped routes is detailed in Supplementary Material Appendix B. Notably, given the availability of the trajectory data, the focus was further narrowed to the subarea containing 2,127 streets with reliable data, ensuring a more robust analysis of the streetscape to answer our research questions on route deviation.

### 3.3. Dependent variable – Desirable bikeshare Index

Inspired by previous studies (Salazar Miranda et al., 2021; Zhao et al., 2024), we developed the Desirable Bikeshare Index to denote the route choice discrepancy compared with shortest routes (formula 1):

$Y_i(DBI) = LIME_i - GOOGLE_i$ (1).

where i represents the ith observation of the road segment, $LIME_i$ the number of actual trajectories traversed, $GOOGLE_i$ the number

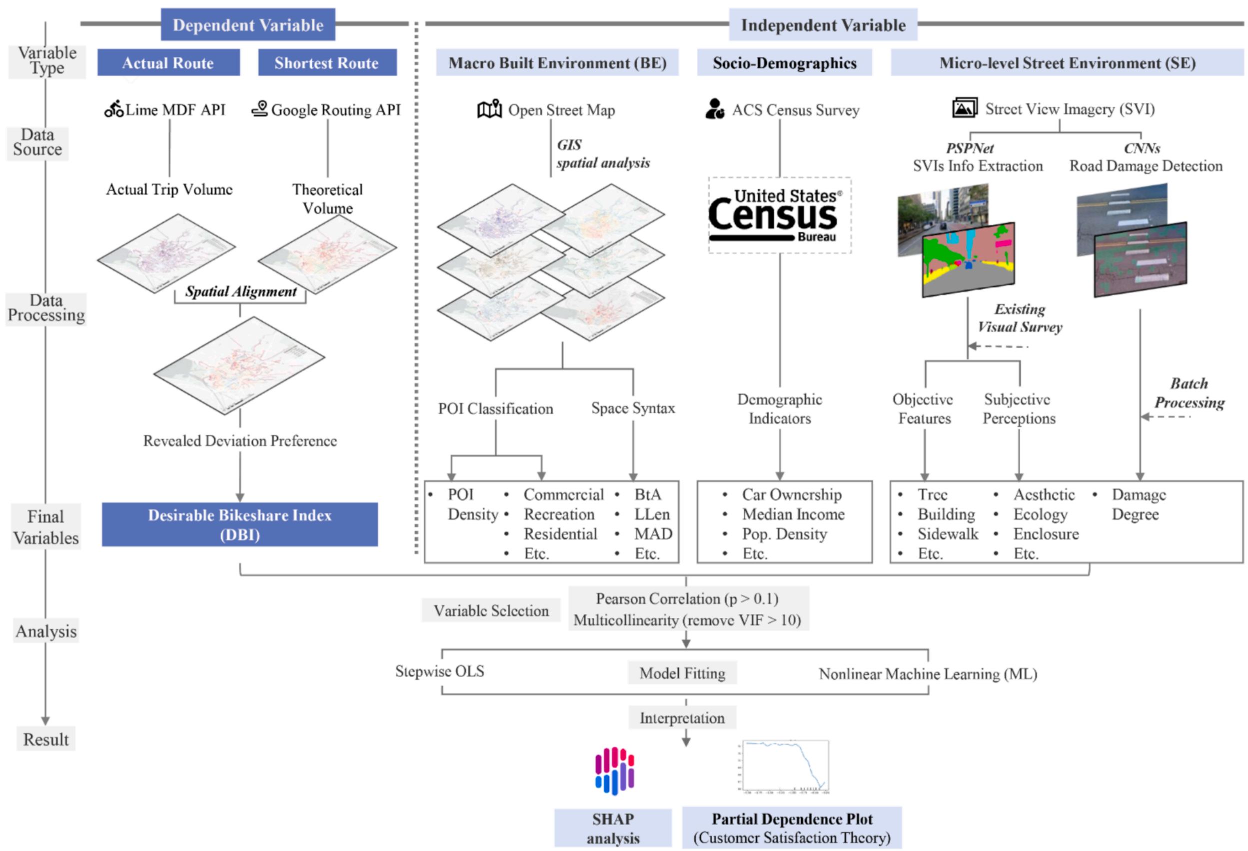



**Fig. 1.** Research Framework. Data & Image.
Source: Street view image samples are from Google Street View API, and basemaps and POI mappings are sourced from Open Street Map (OSM).

of shortest routes traversed.

DBI can mitigate the mechanical effects to isolate the actual impacts of BE and SE attributes – to gauge the varied trip counts of a street segment passed by any actual trip versus its shortest path. The actual and theoretical frequencies of traversals for each uniquely identified segment in the road network were obtained and operationalized to measure its corresponding DBI (Fig. 2).

First, for each trip, the origin–destination (OD) coordinates were fed into the Google Directions API, which returned the shortest path based on the road network. Second, to construct the DBI, a street segment was incremented by one point (+1) when traversed by the actual trajectory, while decremented by one (−1) when only traversed by the simulated shortest route. Therefore, a 0-scored segment suggests that actual and shortest routes are identical for every corresponding OD pair concerning this street. Finally, the cumulative traversal counts for both metrics are combined for each segment, effectively representing the actual street-level DBS deviation preference relative to the shortest routes, though it may underestimate preferences for segments that frequently coincide with shortest routes.

Theoretically, people would minimize travel costs, implying alignment between the shortest and actual routes, i.e., with DBI close to 0. Accordingly, a road segment exhibiting high DBI may suggest specific exogenous attributes that attract DBS cyclists to deviate from the shortest routes for complementary effects by maximizing other unexplained utility apart from time cost (Meyer, 2016), for instance, to experience a visually pleasant streetscape as local responses without hampering the global preferences towards the destinations (Oyama, 2024). Fig. 3 illustrates the aggregated shortest paths, actual routes, and DBI development.

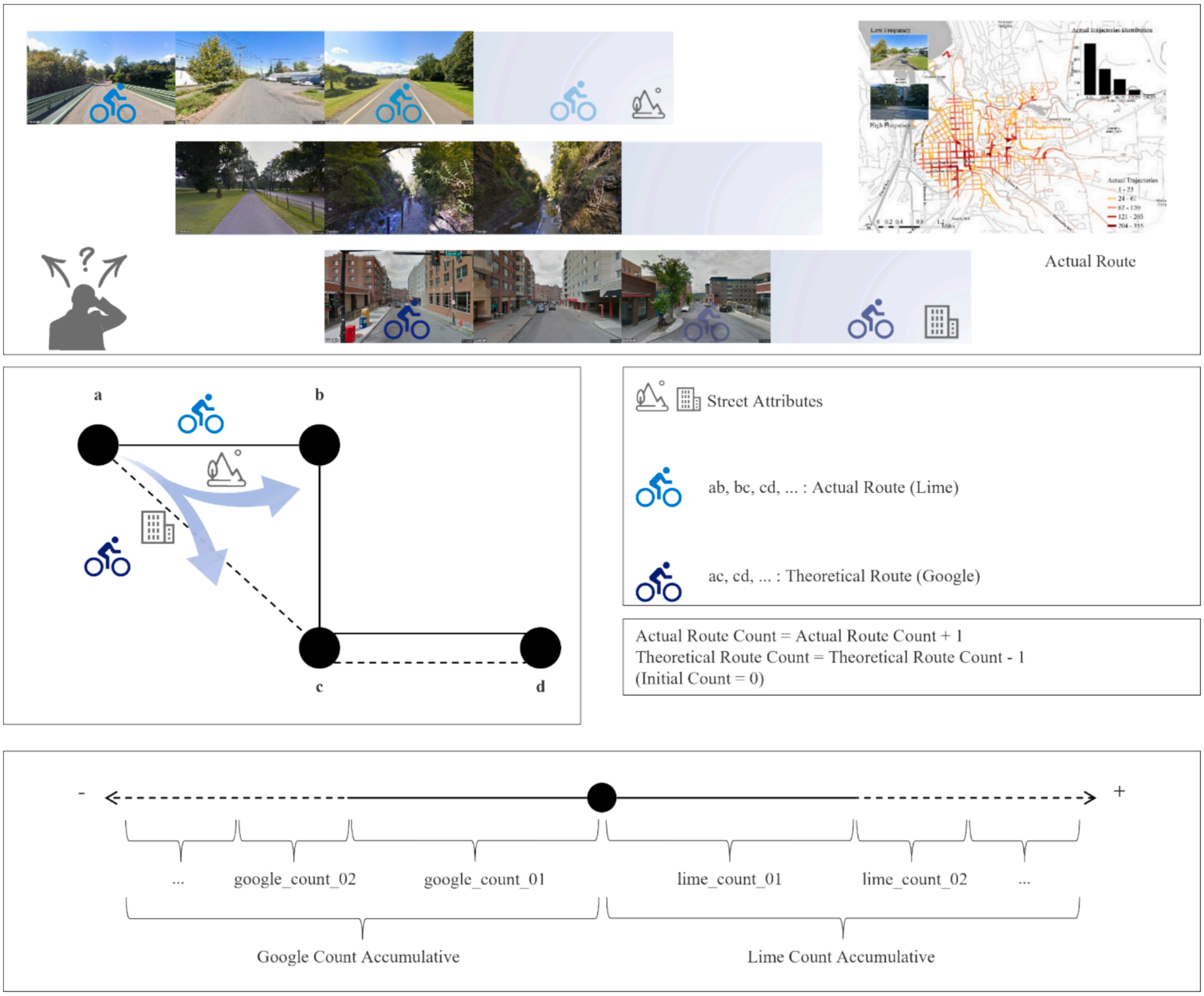
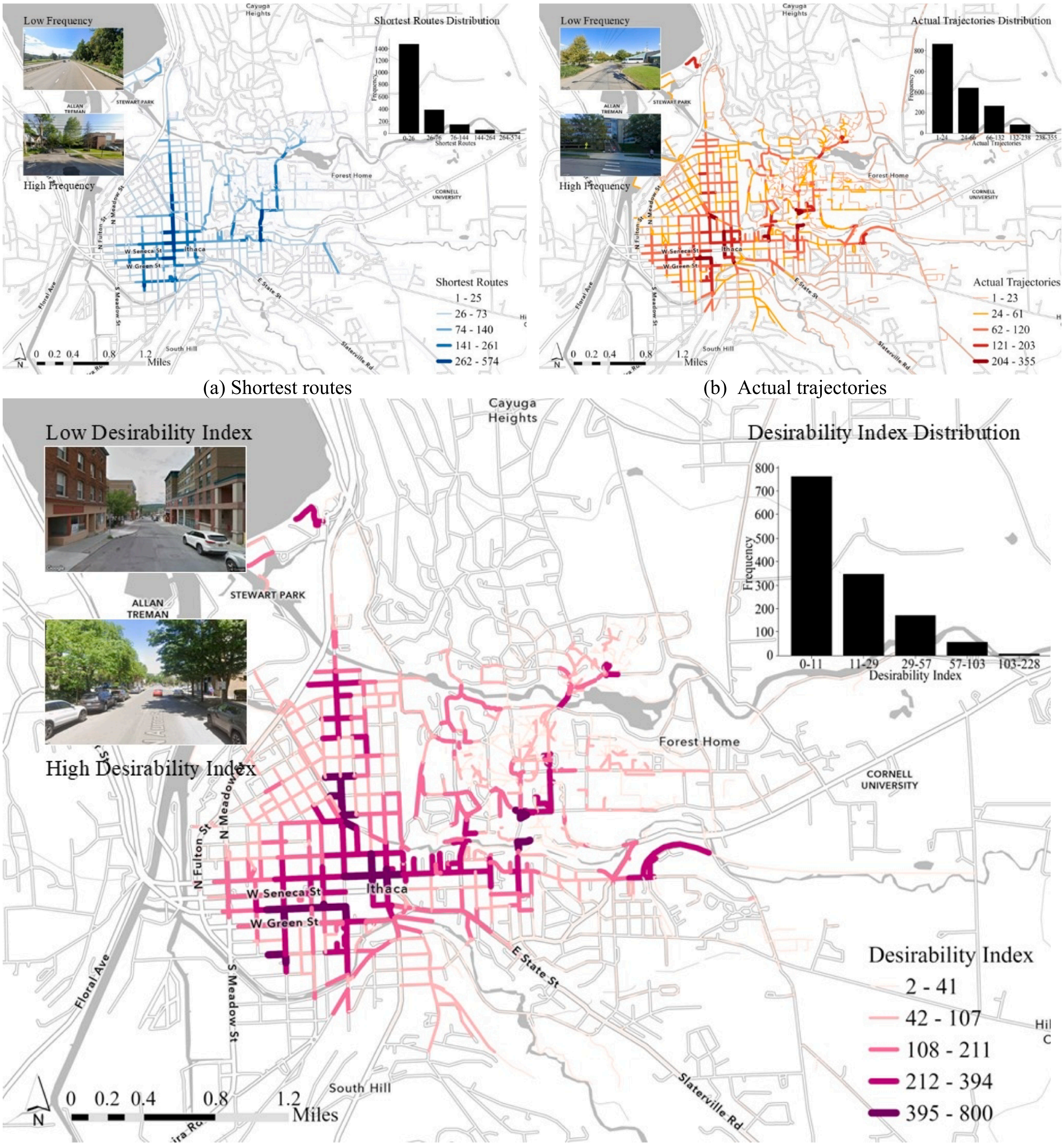



**Fig. 2.** DBI Development.

### 3.4. Independent variables

#### 3.4.1. Micro-level SE attributes

*3.4.1.1. Objective visual features.* To analyze the micro-level SE, we leveraged the SVI downloaded from Google Street View. Extensive studies have used SVI to proxy the street environment experienced by cyclists and to further understand its impact on cycling behaviors (Ito and Biljecki, 2021; Rui and Xu, 2024), with its effectiveness in reflecting ground truth and reliability validated by scholars (Vanwolleghem et al., 2014). Sampled at the 10-meter interval in QGIS, we obtained 11,426 SVIs. Pretrained on the ADE20K dataset, the PSPNet algorithm was employed to segment objective features into view indices. The process is explained in Appendix C.




(a) Shortest routes (b) Actual trajectories

(c) DBI

**Fig. 3.** Mapping aggregated routes: (a) Shortest path, (b) Actual trajectories and (c) DBI.

Additionally, we adopted the Mask R-CNN algorithm to quantify the number of automobiles (ct_traffic) in SVI, reflecting traffic volume; the descriptive summary is in Appendix Table A2.

*3.4.1.2. Subjective perceptions.* Prior studies have not explicitly revealed how subjective perceptions affect cyclists' detours from the shortest routes. Following established research using expert feedback on visual surveys to collect psychological (e.g., lively) (Dubey et al., 2016) and design perceptions (e.g., complexity) using an ML-based framework (Qiu et al., 2022), we enriched the existing knowledge by including seven perceptions: (1) Order, (2) Accessibility, (3) Aesthetics, (4) Ecology, (5) Enclosure, (6) Richness, and (7) Scale. These variables are selected for their ability to capture the micro-level street qualities that influence cyclist route choice. The selection is based on three criteria: (1) actionability, (2) capturing "quality" over "quantity", and (3) cognitive navigation.

Classical urban design frameworks identify fundamental spatial qualities such as 'Richness', 'Enclosure', and 'Scale' (Ewing and Handy, 2009), which are directly tied to physical zoning and design interventions, making the results actionable for planners. In this case, richness refers to the visual complexity of a streetscape − the diversity and variety of features that create a dense, richly detailed scene (Tian et al., 2021). Enclosure is defined as the degree to which vertical elements (e.g., buildings, trees) frame the space (Ewing and Handy, 2009), and Scale refers to the texture, size, and detailing of physical elements that align with human proportions and speed of movement (Ewing et al., 2013; Gehl and Rogers, 2010; Qiu et al., 2023).

Meanwhile, objective view indices such as the Green View Index (Li et al., 2015) quantitatively capture the presence of features but fail to capture their quality or coherence based on recent empirical studies (Wang et al., 2021). Therefore, we added Ecology and Aesthetics, as ecology refers to the perceived natural greenness, and aesthetics refers to the visual beauty and design character of the environment, reflected in color variety, scenic elements, and identifiable aesthetic features (Chen and Biljecki, 2023).

Lastly, for route choice specifically, cyclists rely on visual cues regarding safety and ease of movement. We therefore included Order and Accessibility. Based on previous research, we define order as perceived coherence and organization of spatial elements that contribute to visual clarity and facilitate users' understanding of the environment, and make people involved in the cognitive process of organizing built environment structure information. In the context of cycling, perceived order supports navigation by providing visually legible cues regarding movement and safety (Brettel, 2006; Gjerde, 2011; Handy et al., 2006; Lynch, 1960). Lastly,

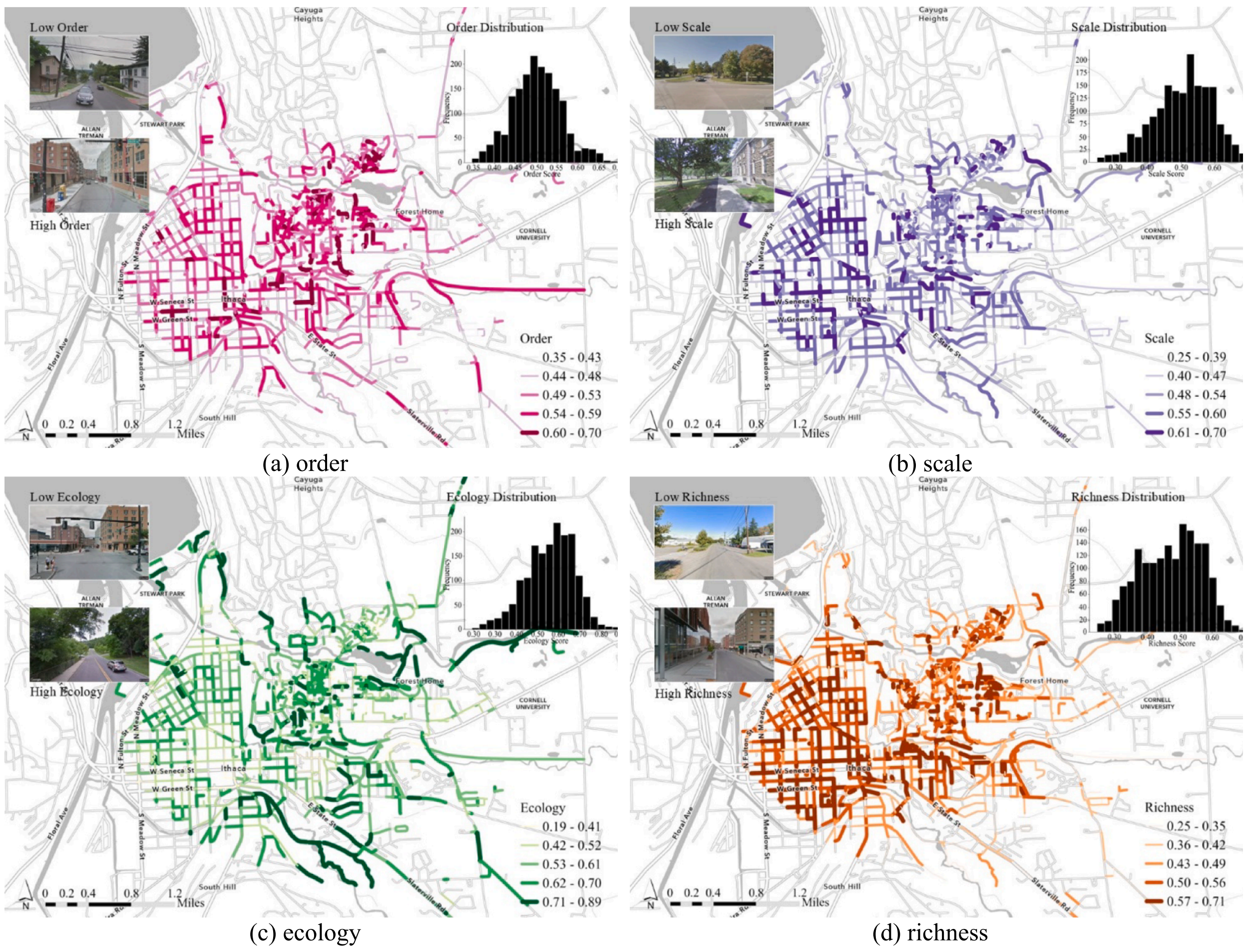



**Fig. 4.** Distribution of subjective perceptions: (a) Order, (b) Scale, (c) Ecology, (d) Richness.

accessibility in our context refers to perceived accessibility, representing individuals' subjective assessment of the ease and convenience of reaching valued destinations, reflecting experiential rather than purely spatial measures, as this process is fundamentally subjective, experiential, and meaning-based (Lättman et al., 2018). In this case, these variables are expected to capture structural organization and visual clarity of the street, which have been shown to serve as proxies for perceived safety and ease of navigation that macro-level road network data cannot fully represent.

Using perception scores from a random sample of 300 SVIs, we trained several ML models and selected the Random Forests algorithm to predict these qualities citywide. For simplicity, the survey methodology, including participant characteristics, pairwise comparison procedures, adoption of the TrueSkill scoring algorithm, assessment against recent methodological standards, and training and prediction processes, is presented in Appendix D. Notably, Fig. 4 illustrates the spatial patterns of four selected perceptions, such as richness and scale.

*3.4.1.3. Quantifying road maintenance conditions.* We acknowledge the pivotal role of road maintenance and its influence on route preferences. However, a scalable method has been lacking to assess such characteristics and control their impact on travel behaviors. Precise identification and delineation of road maintenance can significantly enhance navigation towards safer paths, thereby tangibly improving the overall cycling experience. Leveraging the DL algorithm, we trained a CNN model to classify cracks and fissures in road surfaces in the SVI. The detailed method is described in Appendix E, and after the first epoch, the model achieved 87% training accuracy and 97% validation accuracy, ultimately reaching a validation accuracy of 98.3% after the fifth epoch, signifying its reliability in detecting surface damage (Fig. 5). Although the learning curves exhibit rapid convergence with validation performance saturating after a single epoch and exceeding the corresponding training accuracy, such behavior is consistent with fine-tuning a frozen pretrained backbone, where only a lightweight classification head is optimized. Pretrained CNN backbones such as ResNet50 are known to produce highly discriminative and often nearly linearly separable representations for downstream tasks, enabling very fast optimization of linear heads (He et al., 2016; Kornblith et al., 2019). The higher validation accuracy is also expected, as data augmentation is applied exclusively during training and introduces distributional perturbations absent in validation, while stochastic components such as dropout and batch normalization noise affect only the training phase. In contrast, validation is performed deterministically on clean inputs, yielding more stable estimates (Goodfellow et al., 2016; Szegedy et al., 2016). These properties collectively make the observed convergence pattern reasonable within this transfer-learning setup.

To depict street-level maintenance conditions, we collected SVIs at 10-meter intervals, with camera angles finetuned to capture the road surface through a bird's-eye perspective. Using the CNN model to predict on the acquired 60,363 high-resolution SVIs, we identified cracks in each SVI and their percentage relative to the road surface. Further averaging the crack percentage of SVI by streets, Fig. 6 reveals that several routes in the urban peripheries and along the creek are ill-maintained in Ithaca.

### 3.4.2. Control variables – Macro-level BE and sociodemographic factors

Following prior literature, we included multiple BE and sociodemographic variables to control their effects on route choices (Broach et al., 2012; Cubells et al., 2023; Meister et al., 2023; Prato et al., 2018).

**Accessibility Attributes:** To control the effects of network characteristics (Chou et al., 2023) on detours, we included four indicators of street network accessibility attributes using Space Syntax. Line length (*LLen*) refers to the Euclidean polyline length, while Line Bearing (*LBear*) is the compass bearing between line endpoints, measured from the positive Y direction of the projected grid north. Radius measures, including Betweenness Angular (*Bta800c*) and Mean Angular Distance in Radius (*MAD800c*), capture betweenness and mean angular distance within the 800 m radius (5-minute ride), reflecting the flow and degree of network twists at the

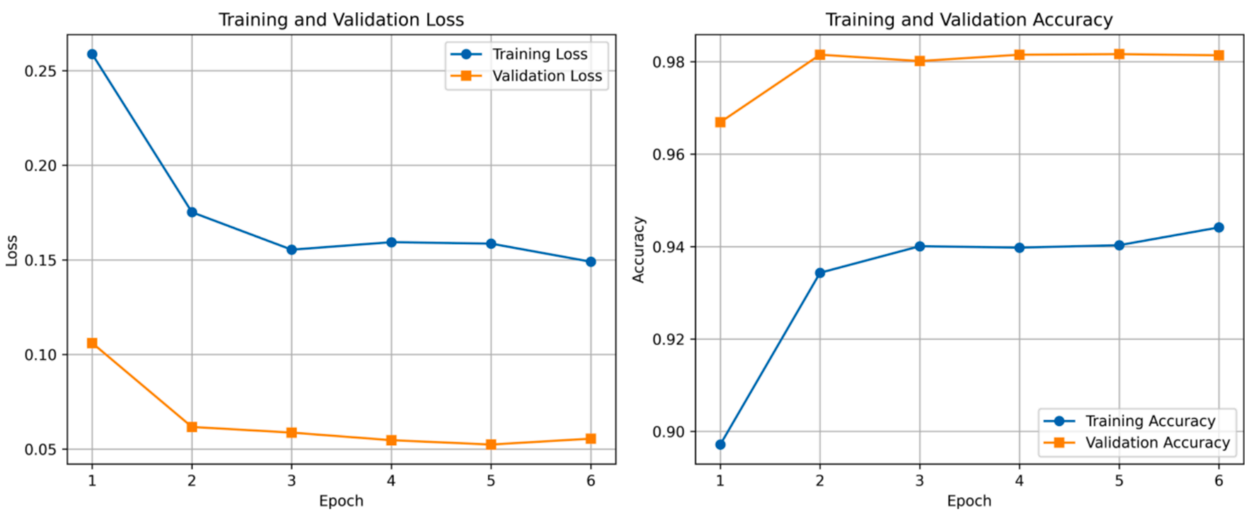
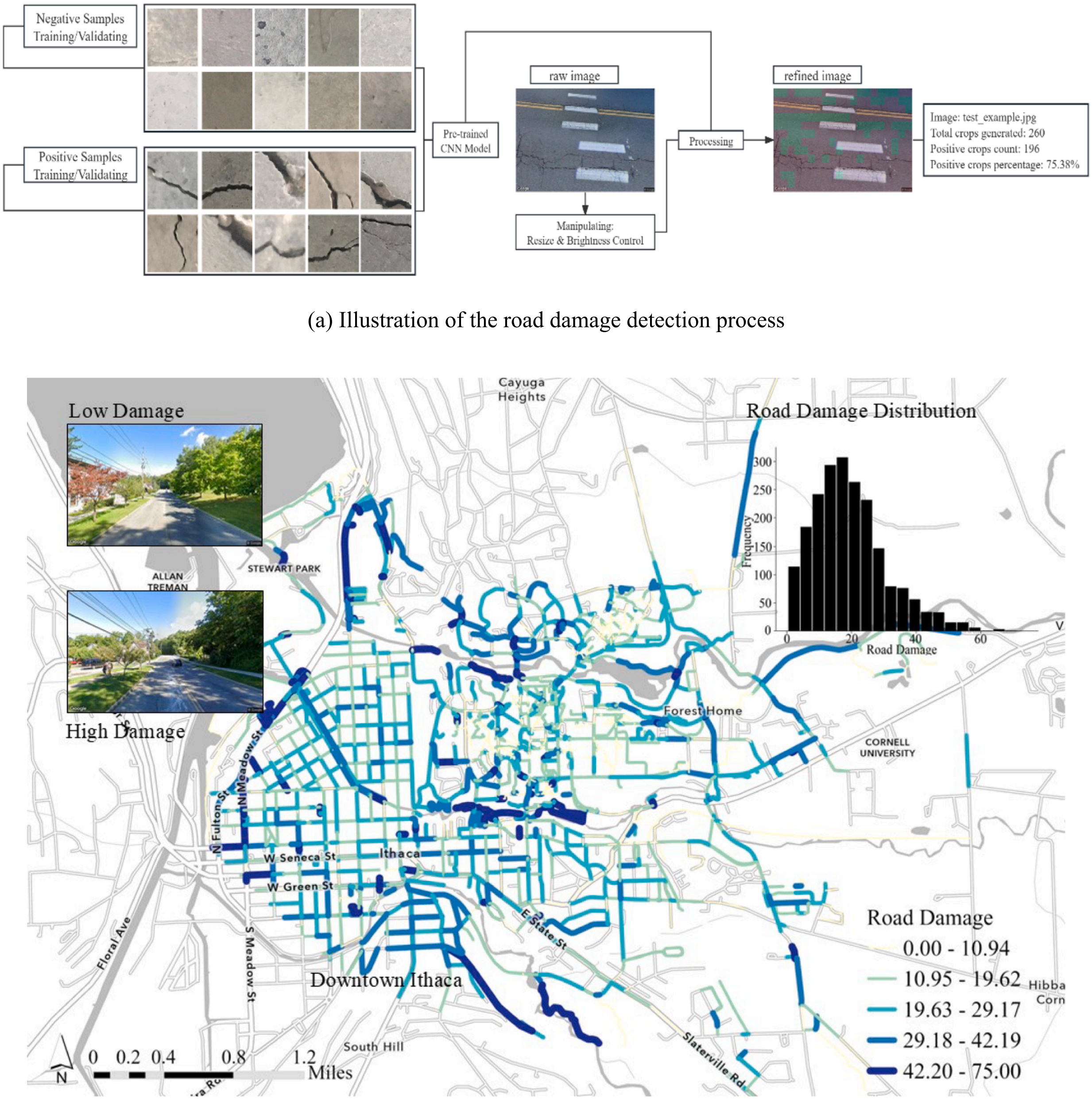



**Fig. 5.** CNN Model Performance. (a) Training and Validation Accuracy, (b) Loss.




(a) Illustration of the road damage detection process

(b) Road Damage Mapping

**Fig. 6.** Road damage detection and results.

neighborhood scale.

**Land use Attributes:** Seven POI types were sourced from OSM: (1) Commercial, (2) Community, (3) Forest, (4) Industrial, (5) Public Services, (6) Recreation, and (7) Residential. The number of POI along each street was determined by tallying occurrences within a 50-meter buffer of each road segment. We also calculated the Recreation-to-Commercial ratio (R_C_ratio) and the total POI count (*POI_Count*).

**Infrastructural Attributes:** Physical characteristics and bike facilities also affect route choices (Fitch and Handy, 2020; Park and Akar, 2019). Therefore, we calculated the slope gradient, provision of bike lane width within a 500 m radius, and road hierarchy, such as whether it is a primary or secondary road.

**Sociodemographic Attributes:** Finally, utilizing the American Community Survey, population density (*PopDen*), work population density (*WorkDen*), and total household (*TotHH*) were obtained at the census block level. Median household income (*Median_inc*) and average vehicle ownership (*vehicle_po*) were collected at the block group level. Each road segment was assigned the value of the nearest or containing census block or block group polygon. The descriptive summary of all control variables is explained in Appendix **Table A2**.

### 3.5. Statistical analysis – OLS regression

In this study, Pearson Correlation was first employed to filter and retain variables demonstrating significant ($p < 0.1$) correlations with the dependent variable. Second, to alleviate multicollinearity among variables, we removed those with high (>10) variance inflation factors (VIFs); for instance, we ultimately retained 12 objective features out of 31 parsed variables. Third, using OLS regression, the macro-level BE, sociodemographic variables, SE objective features, and subjective perceptions were fitted into the DBI separately to reveal the explanatory power of individual attribute groups. Finally, several OLS models were constructed, incrementally adding each attribute group to infer the DBI using the following strategy:

*Model 0 (Baseline): Macro-level BE + Sociodemographic.*
*Model 1: Baseline + Micro-level SE (Objective Features).*
*Model 2: Baseline + Micro-level SE (Subjective Perceptions).*
*Model 3: Baseline + Micro-level SE (Objective Features + Subjective Perceptions).*

### 3.6. Exploring nonlinear relationships using ML

However, the underlying linear assumption of OLS implies that it cannot accurately reflect the potential nonlinear impacts of micro-level SE attributes on route desirability. Therefore, building upon Model 3, the research advanced into ML algorithms. Five mainstream explainable ML models, including Random Forests (RF), Decision Trees (DT), Support Vector Machines (SVM), Gradient Boosting (GB), and XGBoost (XGB), were tested, which are adept at modeling threshold and interaction effects while offering clear interpretation in urban research. The dataset was divided into training (70%), validation (15%), and testing (15%) sets to optimize model performance on out-of-sample data. The best-performed model in the validation set, the RF model, was finetuned to reveal the nonlinear relationship between the built environment and the DBI.

Furthermore, the three-factor framework (basic/excitement/performance) in customer satisfaction studies (Kano et al., 1984) was adopted to discuss the nonlinear impacts and quality thresholds (Ding et al., 2022; Zhang et al., 2025), which were visualized using Partial Dependence Plots in Python. Among the three classified categories, basic elements are essential but do not enhance satisfaction. In contrast, performance elements increase satisfaction proportionally to their quality, and excitement elements, though not essential, significantly boost satisfaction through innovation (Kano et al., 1984). Detailed interpretations of the Kano model are provided in **Appendix F**. The threshold effects of various characteristics on the DBI were carefully examined to address our research questions, yielding insights into shaping a better DBS cycling route experience.

## 4. Results

### 4.1. Strengths of attribute groups

Table 1 shows that the SE variables achieve reasonable goodness-of-fit in predicting DBI, compared with conventional BE factors, while neighborhood-level sociodemographic attributes exhibit marginal influence. The predictive power of attribute groups ranks as follows: Macro-level BE > Micro-level SE (Objective) > Micro-level SE (Subjective) > Sociodemographic. Notably, F-statistic tests for all groups are significant ($p < 0.01$), suggesting that both objective features and subjective perceptions substantially influence the desirability of the DBS route, warranting the integration of micro-level SE to predict the DBI.

### 4.2. OLS performances

Next, multiple OLS models were established to compare their performances (Table 2), using macro-level BE and sociodemographic as the baseline. First, adding micro-level SE attributes to Model 0 significantly improves the explanatory power, with the contribution higher from objective features (M1 by 6.2%) than that of subjective perceptions (M2 by 2.6%) when adding the group separately. Second, Model 3 demonstrates that objective features and subjective perceptions complement each other in influencing route deviation, with the highest predictive strength (36.2% of the variance) and the lowest Akaike Information Criterion (AIC).

**Table 1**
OLS Regression Results.

| OLS Regression | Sociodemographic | Macro-level BE | Micro-level SE (Obj) | Micro-level SE (Sbj) |
|---|---|---|---|---|
| No. of variables | 5 | 18 | 15 | 7 |
| R2 | 0.049 | 0.270 | 0.167 | 0.133 |
| Adjusted R2 | 0.047 | 0.264 | 0.161 | 0.130 |

**Table 2**
Results of OLS models.

| OLS | Model 0 Baseline<br>Macro-level BE + Sociodemographic | Model 1<br>M0 + Micro-level SE (Obj) | Model 2<br>M0 + Micro-level SE (Sbj) | Model 3<br>M0 + Micro-level SE (Obj & Sbj) |
|---|---|---|---|---|
| R2 | 0.278 | 0.314 | 0.304 | 0.339 |
| Adjusted R2 | 0.270 | 0.302 | 0.294 | 0.326 |
| F-statistic | 36.83*** | 26.52*** | 31.54*** | 24.86*** |
| AIC | 24,900 | 24,820 | 24,830 | 24,750 |

Notes: p-values *** $<$ 0.01.

### *4.3. Determining nonlinear relationship*

Notably, comparing various nonlinear ML models with OLS shows that adopting ML algorithms can yield greater explanatory power than linear models. The RF performs better than the other models when $R^2$ and MSE are used as criteria. The parameter settings were obtained by optimizing hyperparameters via 5-fold cross-validation with Grid Search, resulting in the Hyperparameter RF, which achieved the highest R2 of 0.48 (Table 3). The finetuned hyperparameters are as follows: max depth = 30, n_estimators = 400, max_features = 0.3. These parameters balance model fit and generalization capacity by controlling complexity and increasing feature randomness.

### *4.4. Kano result*

Among all variables, perceived accessibility, order, and richness, as well as objective features such as trees, signboards, and vehicle presence serve as excitement factors. Scale, along with buildings, awnings, and road-surface conditions, generally aligns with performance factors, which exhibit approximately proportional increases in desirability as their values rise. In contrast, enclosure, ecology, and aesthetics function as reverse-basic or reverse-excitement factors, in which increases beyond moderate levels reduce cyclists' preference for detours.

Macro-level built-environment attributes also show heterogeneous patterns. Commercial and residential POI densities demonstrate excitement-type thresholds, while population density, work density, total households, community facilities, and space-syntax measures such as MAD800c behave like basic factors with diminishing returns after low–mid thresholds. Bike lanes, line length, forest, and mean slope ultimately exhibit reverse-performance or reverse-excitement patterns, suggesting negative nonlinear associations with detours.

## 5. Discussion

### *5.1. Significant nonlinear effects*

We uncovered significant nonlinear relationships between micro-level SE, macro-level BE variables, and the DBI. Among the top ten variables showing the highest SHAP importance (**Appendix Fig. A1**), perceptions outnumber macro-level BE and other micro-level objective SE variables. The most critical and positive explanatory factors are sidewalks and population density, largely aligned with prior studies (Ghanbari et al., 2024). These findings reveal subtle yet dynamic associations between micro-level SE and route choices that have previously been overlooked (Broach et al., 2012; Prato et al., 2018; Zhao et al., 2024).

### *5.2. Roles of attribute groups*

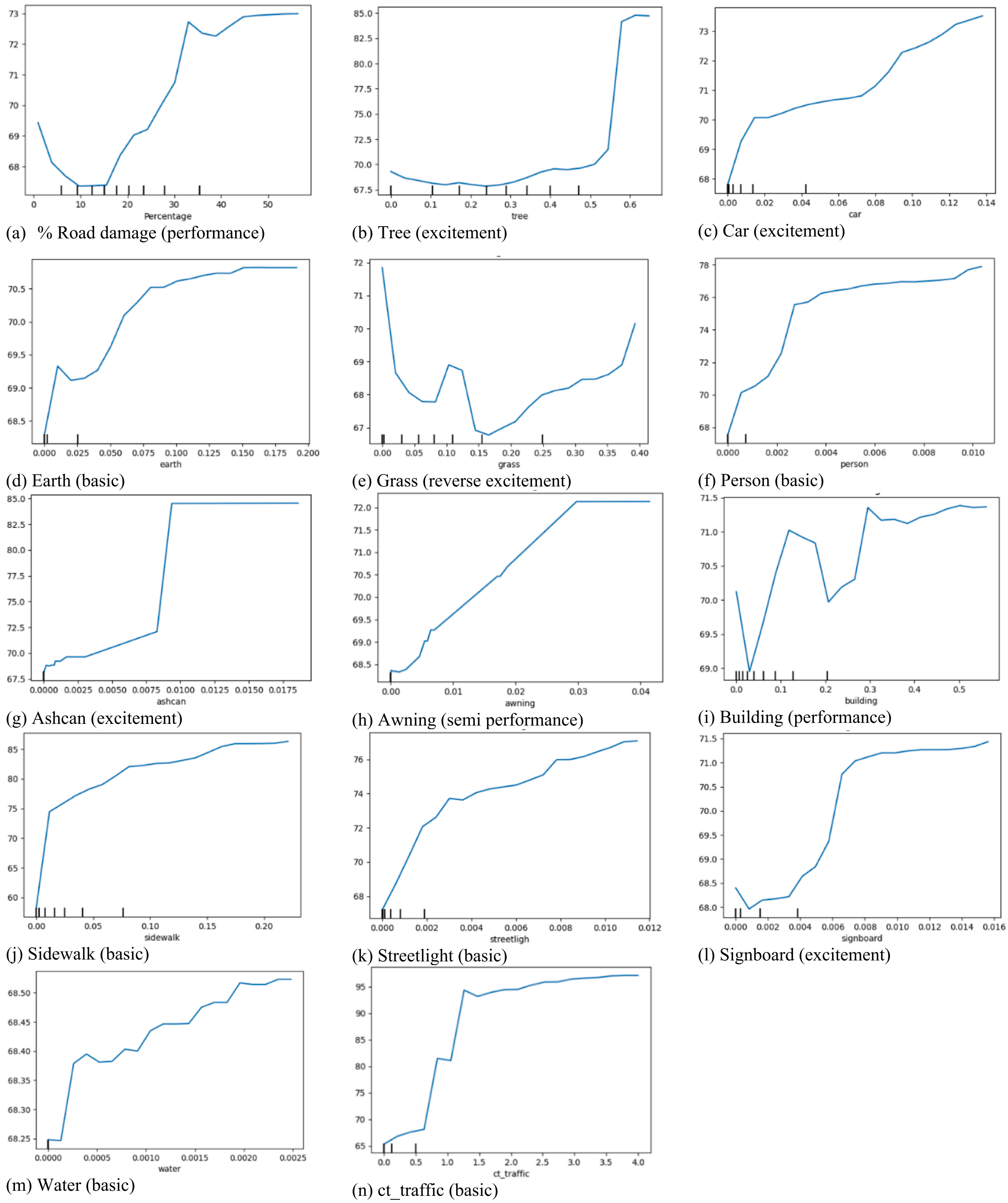
#### *5.2.1. Micro-level objective features*

Most visual features exhibit threshold effects, with trees and grass playing distinct roles (Fig. 7). On the one hand, the tree resembles an excitement factor, with its marginal impact more pronounced after passing the medium threshold, suggesting that trees promote cycling enthusiasm more when they are sufficiently provided (Lusk et al., 2020). A possible explanation is that lush street vegetation provides adequate microclimate control, thereby enhancing cyclists' perceived comfort. Unlike trees, ground cover vegetation (grass) exhibits a more complex relationship with cyclists' satisfaction. Initially, it reduces satisfaction to a low level, then increases it to a moderately high level, thereby establishing two threshold values. This concave relationship adds nuance to prior studies demonstrating the positive impacts of visual greenery on route choices (Lu et al., 2019; Park and Akar, 2019). Our findings

**Table 3**
ML Model Performance.

| Models | OLS | RF | DT | SVR | GB | XGB | Hyperparameter RF |
|---|---|---|---|---|---|---|---|
| R2 | 0.31 | 0.46 | −0.05 | −0.04 | 0.42 | 0.36 | 0.48* |
| MSE | 6202.35 | 4806.26 | 9401.51 | 9336.00 | 5201.55 | 5758.63 | 4679.97 * |
| RMSE | 78.75 | 69.33 | 96.96 | 96.62 | 72.12 | 75.89 | 68.41 |

Notes: * denotes the best performance model.




(a) % Road damage (performance) (b) Tree (excitement) (c) Car (excitement)
(d) Earth (basic) (e) Grass (reverse excitement) (f) Person (basic)
(g) Ashcan (excitement) (h) Awning (semi performance) (i) Building (performance)
(j) Sidewalk (basic) (k) Streetlight (basic) (l) Signboard (excitement)
(m) Water (basic) (n) ct_traffic (basic)

**Fig. 7.** PDP of the objective features.

indicate that supplying minimal ground cover planting does not necessarily promote cycling unless it surpasses a critical threshold (0.37), beyond which satisfaction improves significantly.

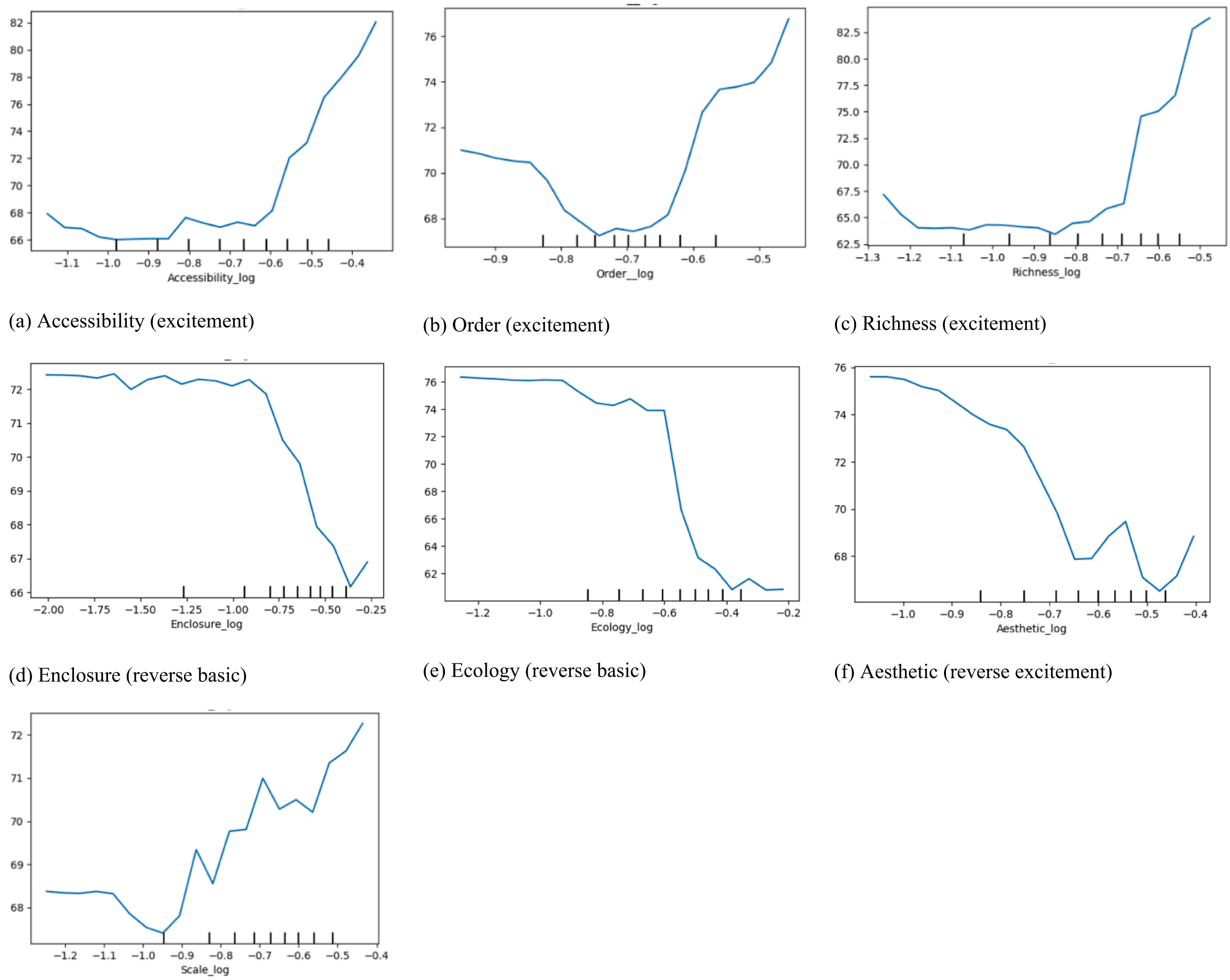
Meanwhile, though road damage follows a performance-factor pattern, the implications differ. Poor road maintenance is associated with cyclists' preferred detour routes, underscoring the urgent need to re-evaluate the existing conditions. Transportation planners should identify detour hotspots with poor maintenance, assess whether the current road network upkeep is adequate for high-traffic areas, adjust maintenance levels as needed, and allocate resources to sustain the high demand for DBS usage.

Furthermore, pedestrian and traffic volumes exhibit different patterns compared to cars (Fig. 7). Concurrently, considerable foot traffic and vehicle volumes appear where more DBS users are present, though this is reasonable and consistent with prior studies (Lu et al., 2018). However, this raises concerns about potential traffic collisions, especially in densely populated areas where overcrowding could eventually deter cyclists from choosing certain routes (Vedel et al., 2017). Moreover, while the vehicle serves as an excitement factor, its co-presence increases the risk of accidents between cars and bikes, suggesting that bikeshare operators and transport planners must prioritize cyclists' safety (Sener et al., 2009).

Lastly, streetlights, water, and signboards demonstrate a positive correlation with route choices, as supported by several studies (Zhao et al., 2024). However, some research reports negative impacts, suggesting that excessive stops can increase time costs (Chou et al., 2023; Cubells et al., 2023). Nonetheless, infrastructure signals are crucial for cyclists and often positively influence route choices (Fitch and Handy, 2020; Meister et al., 2023). Additionally, buildings and awnings positively relate to DBI and serve as performance factors, contradicting previous research, which found that urban centers with higher density may attract fewer cyclists (Winters et al., 2010). However, the complicated role of perceived enclosure (discussed later) may provide more nuanced insights, as relying solely on sky and building views often yields mixed findings.

### 5.2.2. *Micro-level subjective perceptions*

Navigation algorithms often assume people make rational route choices by minimizing time costs (Meyer, 2016), overlooking users' subjective preferences for local route characteristics (Meng and Zheng, 2023). Our results indicate that macro-level BE features alone cannot fully explain variations in DBS route choices; instead, such desirability is further elucidated through subjective perceptions




(a) Accessibility (excitement)

(b) Order (excitement)

(c) Richness (excitement)

(d) Enclosure (reverse basic)

(e) Ecology (reverse basic)

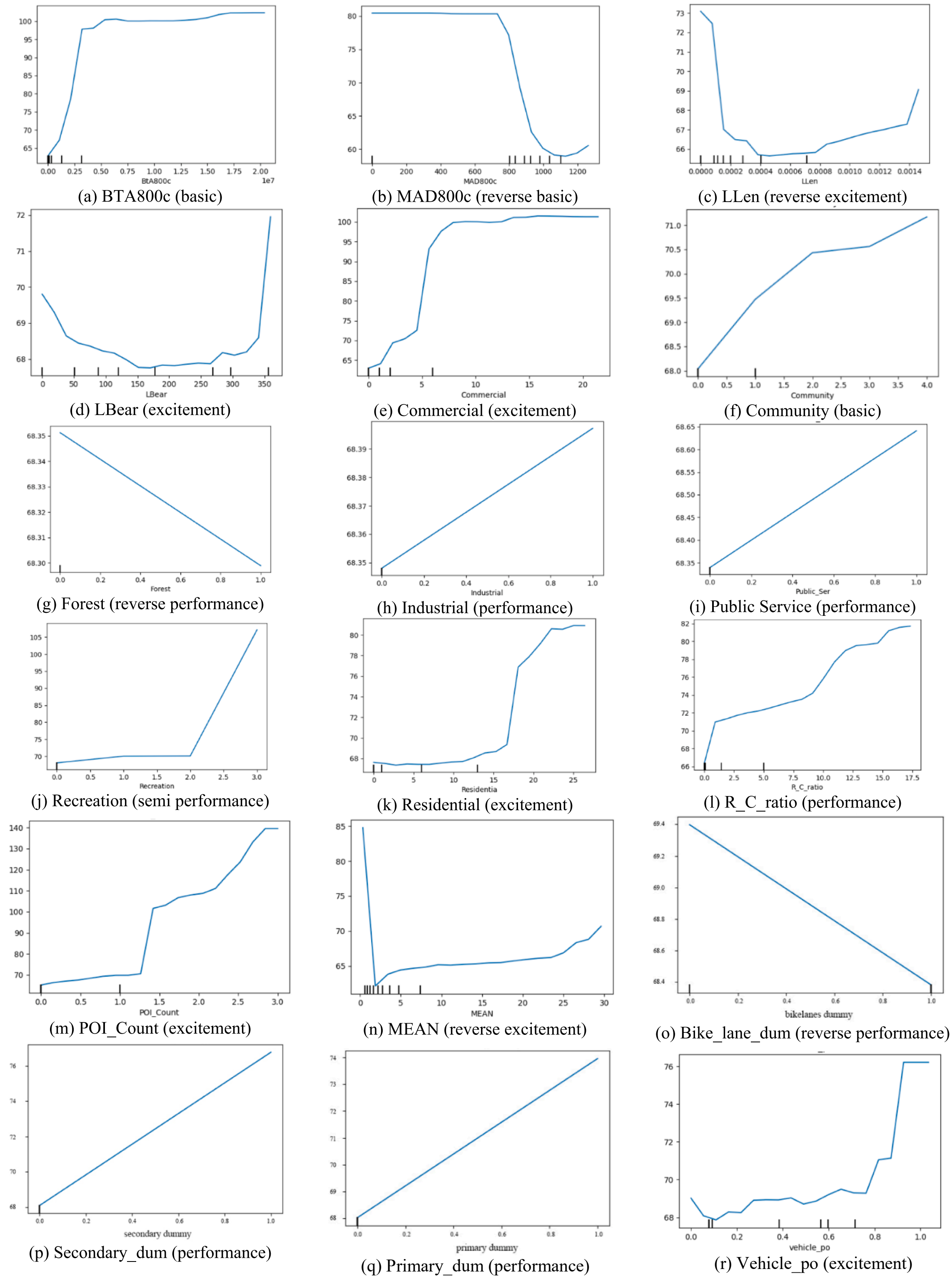
(f) Aesthetic (reverse excitement)

(g) Scale (performance)

**Fig. 8.** PDP of the Subjective Perceptions.

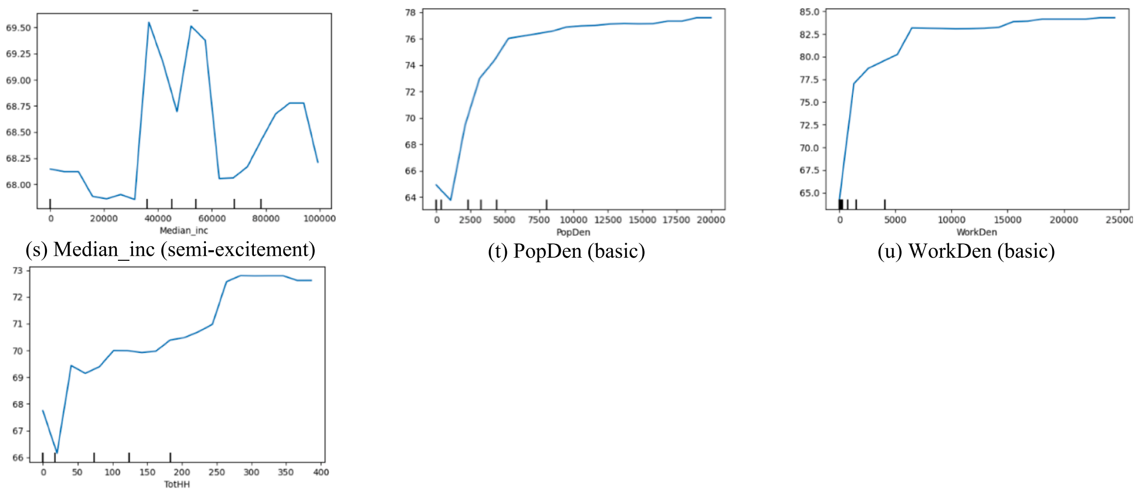
(Blitz, 2021; Fernández-Heredia et al., 2014; Rui and Xu, 2024).

First, perceived accessibility, order, and richness constitute key excitement factors. For accessibility (Fig. 8a), streets with seamless access to bike facilities significantly attract more users who deviate to such routes, consistent with the literature (Guo and He, 2021).




(a) BTA800c (basic) (b) MAD800c (reverse basic) (c) LLen (reverse excitement)
(d) LBear (excitement) (e) Commercial (excitement) (f) Community (basic)
(g) Forest (reverse performance) (h) Industrial (performance) (i) Public Service (performance)
(j) Recreation (semi performance) (k) Residential (excitement) (l) R_C_ratio (performance)
(m) POI_Count (excitement) (n) MEAN (reverse excitement) (o) Bike_lane_dum (reverse performance)
(p) Secondary_dum (performance) (q) Primary_dum (performance) (r) Vehicle_po (excitement)

**Fig. 9.** PDP of the macro-level BE and sociodemographic variables.




(s) Median_inc (semi-excitement)
(t) PopDen (basic)
(u) WorkDen (basic)
(v) TotHH (basic)

**Fig. 9.** (*continued*).

Notably, its initial marginal effect increases substantially after surpassing the medium threshold (Zhang and Hu, 2024). Similar trends are observed in order (Fig. 8b) and richness (Fig. 8c), suggesting that DBS cyclists prefer visually complex and less chaotic routes over shortest paths when these criteria reach mid-to-high levels. These results align with research suggesting that complexity triggers visual interest and enhances quality perceptions (Blitz, 2021). However, they contrast with Salazar Miranda et al. (2021), who found a negative correlation for pedestrian routes, likely due to differences in mode. Furthermore, the scale can be viewed as a performance factor (Fig. 8g), echoing prior findings that human-scale streets encourage more cycling trips (Song et al., 2024).

Interestingly, enclosure, ecology, and aesthetics (Fig. 8d-f) exhibit negative associations with DBI. Notably, visual enclosure follows a “reverse-basic” logic: while medium enclosure may enhance safety, excessive levels deter cyclists. The negative direction aligns with Park and Akar (2019) and Song et al. (2024), while contrasting studies reveal positive effects (Salazar Miranda et al., 2021). Our research enriches existing knowledge, and its reverse trend may corroborate previous inconclusive inferences when analyzed under linear relationship assumptions. A reasonable enclosure can enhance safety, but it overshadows such a preference when it is taken too far. Similarly, consistent with prior literature (Rui and Xu, 2024), perceived ecology is a reverse-basic factor. Though scholars reported that higher perceived ecological quality leads to increased perceived safety (Ferretti et al., 2019), it is reasonable that DBS users may choose familiar routes in the urban areas that are naturally associated with lower ecological quality (Harms et al., 2021). Lastly, the reverse-excitement aesthetic pattern draws contrary evidence to that of Rui and Xu (2024), who found an upward trend. Notably, the apparent U-shape across thresholds underscores the complex mechanism by which DBS route choices are influenced nonlinearly, warranting further validation across diverse contexts.

### *5.2.3. Macro-level BE characteristics*

Our research has shown that macro-level BE variables have significant nonlinear effects on route deviation, indicating that these factors can lead to varied choices depending on specific circumstances (Fig. 9). Space syntax factors play distinct roles in shaping cyclists' route choices. Line bearing is an 'excitement' factor that enhances cyclists' experiences, and mean angular distance is a 'reverse basic' factor. At the same time, betweenness centrality acts as a 'basic' factor, and line length emerges as a 'reverse excitement' factor. This highlights the crucial role of urban network morphology in shaping the cycling experience, with significant implications for urban planning and infrastructure development.

Our research has revealed unexpected findings, including a negative correlation between bike lanes and cyclists’ detours. It contradicts many prior studies (Broach et al., 2012; Fitch and Handy, 2020; Meister et al., 2023) but is supported by a few others (Chou et al., 2023; Chung et al., 2024), underscoring the intricate nature of urban cycling. While this negative association may also partly reflect the DBI's structural underestimation of preferences for high-centrality streets where bike lanes tend to concentrate, the positive effects of primary and secondary road types suggest that wider roads may not pose significant safety concerns in a smaller city with tolerable traffic. However, the fact that cyclists may have fewer alternative routes with dedicated bike lanes highlights the need for future improvements in bike lane infrastructure to enhance the overall cycling experience.

Our research has also unveiled novel findings regarding the impact of land use on the DBI. Specifically, we identified a precise threshold of 7 for commercial POI (Fig. 9e), suggesting that it exhibits noticeable effects at substantial levels of development. Similarly, it exhibits a threshold effect at 17.5 for residential developments (Fig. 9k), beyond which DBI increases sharply, aligning with prior research indicating a positive impact on residential areas (Prato et al., 2018). It is worth noting that although previous studies reported a positive correlation between land use density and route choices (Park and Akar, 2019; Zhao et al., 2024), our findings reveal that these effects are marginal until the threshold is reached. This suggests that cyclists will likely deviate from the shortest paths if the attractions sufficiently outweigh the time costs. Finally, recreational facilities can alter decision-making, with an observable threshold effect at 2 (Fig. 9j). This may be due to the aesthetics (Blitz, 2021) and street greenness (Park and Akar, 2019) associated with green open spaces, which provide valuable visual appeal to cyclists.

#### 5.2.4. Sociodemographic factors

All sociodemographic variables–population density, work population density, total households, and vehicle ownership, except for median income − are positively associated with DBI (Fig. 9r-v). The first three variables serve as basic factors, suggesting that a slight increase below the threshold can lead to a noticeable change in cyclists' detours from the shortest paths. These findings are consistent with previous research indicating that higher population density is associated with greater numbers of commercial facilities, job opportunities, and other environmental factors that can promote active travel (Ewing and Cervero, 2010). Meanwhile, once a certain density threshold is reached, cyclists tend to rely more on personal experience, opting for routes they deem more suitable rather than strictly adhering to the shortest pathways, which can explain the flattened curve.

The median income notably influences the DBI, with significant fluctuations observed at several income levels. A pronounced spike in DBI occurs when the median income reaches thresholds of $40 K and $60 K, peaking at approximately 69.5 routes, then bouncing back and leveling off at 68.5 before dropping again. This surge suggests that higher neighborhood income may be associated with residents choosing longer or more appealing commuting routes in such areas, consistent with a study on behavioral detour ratios in Copenhagen (Chou et al., 2023). Enhanced scenic value, increased safety, or greater amenity availability in high-income neighborhoods can explain such an increase. However, its marginal utility can diminish again when time efficiency becomes more salient for residents (Lu et al., 2023).

### 5.3. Planning implications

Our analysis has several planning implications. First, enhancing street design characteristics that promote in-situ SE perceptions can significantly attract DBS users to detour from the shortest routes. Analyzing existing SE conditions and their effects on route choices provides a viable foundation for planners to model cycling flows through proper calibration at the city scale, thereby deriving active travel demand similar to that observed in walking behavior studies (Sevtsuk, 2021). Second, given the inherent complexities of SE attributes, coupling hotspots of several perceptions with conventional road attributes, such as bike infrastructure, can yield practical insights for pinpointing locations that maximize utility to promote cycling. Notably, it requires the expertise of planners and designers to balance micro-level characteristics and jointly conceive new human-centric street typologies that do not impede existing road functions. Furthermore, we provide a rough cost estimate in Appendix G for the common features that can be considered in practice to shape a cyclist-friendly street interface. Lastly, it provides an excellent foundation for integrating micro-level characteristics into the design of a route-planning platform to facilitate cycling behaviors (Su et al., 2010). Users can explicitly adjust weights based on their heterogeneous preferences for exposure to experiential qualities and environmental features to generate routes iteratively between OD pairs, thereby uncovering underexplored place identities in the city.

Nevertheless, specific empirical findings should be interpreted cautiously when extrapolated beyond the study area, while the broader analytical approach remains widely applicable. Our framework in developing the DBI, the integration of objective and subjective streetscape attributes, and interpretable machine-learning models is transferable for examining how micro-scale environments influence cycling route choices across diverse urban settings. Notably, the exact direction and magnitude of effects are likely to vary with local context and travel mode. For instance, our reverse-basic relationship for enclosure diverges from the generally positive association reported in pedestrian studies (Salazar Miranda et al., 2021). Consequently, we recommend that practitioners conduct small-scale validations or pilot studies to assess the relevance of these factors to local DBS patterns before applying our findings more broadly.

Our research suggests the importance of institutional collaboration where local authorities and private DBS operators can play complementary and mutually reinforcing roles in shaping cycling conditions. Extensive evidence shows that government-controlled factors, such as station placement, cycling infrastructure quality, street connectivity, and land-use planning, are able to directly shape DBS trip distributions, imbalances, and the feasibility of first-/last-mile movements to support public transportation (Cheng et al., 2022; Gao et al., 2023; Wang et al., 2016; Zuo et al., 2020).

Private operators, by contrast, govern day-to-day fleet deployment, rebalancing, and service provision. Their high-resolution mobility data can reveal infrastructure deficits, emerging demand hotspots, and operational issues that are often invisible to conventional planning datasets (Gao et al., 2023). At the same time, the introduction or expansion of metro services substantially reshapes bikeshare usage patterns, underscoring the need for joint planning of station areas and multimodal interchange facilities to support DBS's role as an access–egress mode (Böcker et al., 2020; Gu et al., 2019).

These complementary capacities create opportunities for coordinated decision-making. Private-sector data can help public authorities prioritize investments in protected lanes, pavement enhancements, and spatial-equity interventions using available resources, while enabling operators to refine bike allocation, extend service to underserved communities, and improve operational efficiency. Misalignment between infrastructure provision and fleet deployment can generate inefficiencies and competition, reinforcing the need for a planning platform between planning authorities and DBS operators (Böcker et al., 2020; Gu et al., 2019).

### 5.4. Limitations and future work

Several research limitations should be acknowledged. First, although the study considered a range of BE variables across scales, it neglected temporal dynamics or outdoor comforts, which play essential roles in route choices. For example, the temporal differences between weekdays and weekends, as well as the hourly variations, could significantly affect environmental perceptions, influencing local route choices (Basu and Sevtsuk, 2022). Collecting DBS trajectories over an extended period, along with fine-grained outdoor comfort and microclimate data, can significantly contribute to this gap.

Second, methodologically, we acknowledge that inverse reinforcement learning could serve as a complementary approach for recovering latent behavioral rewards if larger volumes of high-resolution trajectory data become available. Future work can integrate IRL-based models to examine whether behaviorally derived reward functions yield consistent implications with our interpretable environmental models.

Third, although the street-level aggregation strategy adopted in this study is particularly well-suited for identifying infrastructure priorities, our analytical framework does not enable formal causal inference. Although methods such as propensity score matching and generalized random forests are applicable to cross-sectional settings, the fundamental challenges of reverse causality and unobserved confounders in our context cannot be addressed solely by these approaches (Ito et al., 2024). We therefore position our analysis and interpret our results as robust associational patterns reflecting revealed detour preferences rather than causal effects.

Fourth, we acknowledge that there is a structural limitation of the DBI. By construction, street segments where actual and shortest paths coincide contribute zero, implying that attributes concentrated on high-centrality streets, such as bike lanes and line length, may have their positive effects underestimated. Our findings should therefore be interpreted as reflecting detour-based route preferences and should not be simply generalized to cycling preferences in their entirety.

Finally, this study has been limited to a small city in the US with a northern climate. Despite this geographic specificity, our findings demonstrate strong generalizability to cities that share similar characteristics with Ithaca, particularly small and medium-sized cities with emerging DBS systems and bicycle infrastructure under similar climatic conditions. In these contexts, our identification of micro-level street environmental effects, including subjective perceptions (such as accessibility, order, and scale), excitement and performance factors, along with the nonlinear threshold effects of objective features (such as trees, road surface quality, and buildings), provides preliminary actionable insights for bicycle infrastructure planning.

## 6. Conclusion

The transportation literature has predominantly focused on modeling cyclists' route choices and their association with bike facilities and trip characteristics (Meister et al., 2023; Menghini et al., 2010; Prato et al., 2018). In contrast, planning scholars have concentrated more on the effects of macro-level BE factors (e.g., land use) and, more recently, on micro-level SE attributes, such as street greenery (Cubells et al., 2023; Park and Akar, 2019; Zhao et al., 2024). Nevertheless, a notable gap remains in understanding the effect of micro-level SE on route detours from shortest paths, which can directly reflect DBS users' revealed deviation preferences. Adopting an interdisciplinary perspective, our study addressed this gap by presenting a comprehensive analytical framework to construct a novel DBI metric that proxies aggregated detour choices at the street level, using over 4,400 GPS trajectories in Ithaca. We subsequently revealed significant nonlinear threshold effects of micro-level SE − encompassing objective features and subjective perceptions − on route deviations.

In particular, subjective perceptions such as perceived order, accessibility, and richness emerge as key excitement factors, while scale functions as a performance factor, all of which positively affect DBI. Conversely, enclosure and ecology exhibit a reverse basic trend, while aesthetics display a reverse excitement pattern, indicating a more complex negative relationship with route deviation. Among objective features, trees and signboards exhibit an excitement pattern, leading to a marked increase in detours once the threshold is exceeded. Buildings, awnings, and road damage percentage correlate positively with DBI, indicating that they serve as performance factors that underscore the importance of addressing existing street maintenance issues. Additionally, macro-level BE attributes, such as commercial POIs, exhibit nonlinear threshold effects, further enriching the prior empirical literature on route choice behavior. Given the significant nonlinear impacts of micro-level SE on DBS route deviations, the findings of this study provide valuable insights for government officials, scholars, and practitioners. These insights can inform SE redevelopment decisions that shape cycling route choices, ultimately facilitating DBS cycling as a healthy, environmentally friendly active travel mode.

### CRediT authorship contribution statement

**Yuxuan Cai:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Qiwei Song:** Writing – original draft, Validation, Supervision, Project administration, Writing – review & editing, Visualization. **Yiming Cheng:** Writing – original draft, Visualization, Formal analysis, Conceptualization. **Anzhi Chen:** Writing – original draft, Visualization. **Yuankai Wang:** Writing – original draft, Validation, Supervision. **Wenjing Li:** Supervision, Methodology, Data curation. **Waishan Qiu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

This research has been supported by The University of Hong Kong's HKU-100 Scholars Award (recipient Dr. Waishan Qiu). During its data collection phase, this study has also received support from the Alan Black Transportation Related Grant (2020) from the - Department of City and Regional Planning and the Serve in Place Fund (2020) from the Office of Engagement Initiatives at Cornell University (recipient Dr. Waishan Qiu). The authors would also like to thank the City of Ithaca, Bike Walk Tompkins, Lime, and Mr. Hector Chang for providing bikeshare trip data during the summer of 2020. This research could not have been made without their support.

## Appendix A

**Table A1**
Summary of selected literature.

| **Prior Study *(Region)*** | **Active Travel *(Sample Size + Time)*** | **GPS Route VS** Routes for comparison | **Independent Variables *(+/- pos/neg)*** Macro-level BE | Micro-level SE | Socio-demographics | **Dependent Variable(s) /Model *(Accuracy)*** | **Limitations** |
|---|---|---|---|---|---|---|---|
| Walking | | | | | | | |
| Salazar Miranda et al., 2021 *(Boston, US)* | Walking *(120,910 trips); 2014.05–2015.05* | Shortest path | (+) Business; (+) Sinuosity; (+) Slope; (+) Route Service Access; (+) Park | (−) Façade; (−) Complexity; (+) Furniture; (+) Sidewalks; (+) Enclosure | N/A | Desirable Index, Multinomial logit (MNL). *(R2 = 0.05)* | • Poor explainability<br>• Only examined objective environment. |
| Sevtsuk et al., 2021 (San Francisco, USA) | Walking (14,760 trajectories)*; 2014.05–2015.05* | Alternative routes | (−) Length;(−) Turns;(+) Sidewalk width; (−) Traffic volume | Sky View Factor; Green View Index | N/A | Route choice, PSL. (R2 0.679) | Limited objective features.<br>Ignored perceptions. |
| Basu & Sevtsuk, 2022 (Boston, USA) | Walking (11,165 trajectories), 2014 | Shortest path | (−) Turns; (+) Amenities; (+) Sidewalk width | Sky View Factor; Green View Index | N/A | Route choice preferences.PSL (R2 0.825) | • Lacking research of preferences of marginalized groups. |
| Personal bikes from recruited participants / Crowd Sourced GPS devices | | | | | | | |
| Menghini et al., 2010 *(Zurich, Switzerland)* | Bicycling *(3,387 trips –7 days in 2004)* | Alternative routes | (−) Length, (+) speed; (+) Bike Path; (+) Lights; (+) Path Length | N/A | N/A | Route choice, MNL *(R2 0.280)* | • Neglects street network and micro-level SE |
| Broach et al., 2012 *(Portland, USA)* | Bicycling *(1,449 trips, 164 bicyclists) 2007.03–11* | Shortest paths + choice sets generated | (−) Distance; (+) Path Size; (−) Turns; (−) Upslope; (−) Signal; (+) Boulevard (+) Bike Path/Lane; (+) Bridge | N/A | N/A | Route Choice.Path size logit (PSL) . *(R2 0.256)* | • Weather and trip purpose are entered by cyclists before starting trips |
| Prato et al., 2018 *(Copenhagen, Denmark)* | *3,384 trips; 2012.10–12; 2013.06–07; 2013.08–10* | Alternative routes | (+) Direction; (+) Distance; (−) Bicycle Bridge; (+) Paved (+) Residential; (+) Elevation | N/A | N/A | Generalized mixed logit model. *(R2 0.330)* | • Overlooks potential biases in the sample selection process.; Neglects SE |
| Park and Akar, 2019 (Columbus, USA) | 1531 trips from 78 cyclists; 2016.08–11 | Shortest route | (+) Distance; (+) Upslope (+) Land Use; (+) Bicycle Facility | (+) Street Greenness | N/A | Normalized degree of deviation; Multilevel mixed-effect linear regression (MMELR); (log-likelihood = -818) | • Overlooks personal traits like bicyclists' preferences.<br>• individual perceptions |
| Fitch and Handy, 2020 (Davis + San Francisco, USA) | SF (GPS): 8070 trajectories, 2009.11—10.03 / 2013.11–14.03 | Alternative routes | (+) Bike Lane; (+) Off-street Paths; (−) Mixed Traffic on Local Roads; (−) Upslope (+) Signals; (−) Stop signs | N/A | N/A | Route Choice. MNL | • Neglects SE characteristics |
| Meister et al., 2023(Zurich, Switzerland) | 4,432 trajectories, 100 individuals. 2021 | Alternative routes by link elimination | (−) Slope; (+) Traffic signals (−) Length; (+) Bike Lane/Path | N/A | (+) Gender (−) Age (−) E-bike | Route Choice. Mixed PSL (Log Likelihood = -10 589) | • Mixed Logit model's extensive parameterization |

(continued on next page)

**Table A1** (*continued*)

| Prior Study *(Region)* | Active Travel *(Sample Size + Time)* | GPS Route VS Routes for comparison | Independent Variables *(+/- pos/neg)* Macro-level BE | Micro-level SE | Socio-demographics | Dependent Variable(s) /Model *(Accuracy)* | Limitations |
|---|---|---|---|---|---|---|---|
| Chou et al., 2023 (Copenhagen, Denmark) | 287,978 processed Hövding trips; 2019.09 – 2021.05 | Shortest path | (+) Speed Limit; (−) Max. traffic (+) Distance (−) Cycle Superhighway; (−) Traffic light; (−) Network Length (−) Bike Lane; (+) Straightness | **N/A** | (+) Income Level | Detour Ratios. Spatial error model. (R2 = 0.64–0.72) | may lead to overfitting. • overrepresentation of frequent cyclists. • Neglects land use + SE. |
| de Jong et al., 2023 (Oslo, Norway) | 25,915 trips, 2017 | Shortest path | (−) Population Density; (−) Workplace Density; (−) Highway | (−) Green Space (+) Water | **N/A** | Route Choice. Linear regression. *(R2 = 0.101)* | • Overall model fit is weak. • Neglect subjective perceptions |
| Docked Bikeshare / docked system with certain flexibility | | | | | | | |
| Lu et al., 2018 (Hamilton, ON, Canada) | 161,426 trips; 2015.05—2016.05 | Shortest path | (−) Slope;(−) Turn;(−) Length; (−) Intersections; (+) Bike Facilities; (+) Traffic; (+) Route Directness Index; (+) Bike Lane | **N/A** | **N/A** | Route Choice. Paired *t*-test. | • Lacking time-based shortest path alternatives. |
| Khatri et al., 2016 | 9,101 trips, 2014.11 –2015.05 | Alternative routes | (−) Turn; (−) Daily Traffic; (−) Distance;(+) Bike Facilities (−) One-way Segments; (+) Signalized Intersections | **N/A** | **N/A** | Route Choice. PSL | • No information was available on demographic factors. |
| Cubells et al., 2023 (Copenhagen, Denmark) | 696 trips; 63 bike sharing cyclistsOne week in 2021 | Shortest path | (+) Upslope; (+) Intersections | (−) Streetlights; (+) Crosswalk; (−) Stops and Traffic Lights; (+) Bicycle Sharrow; (−) Trees | (+) Male | Detour Percentage. MMELR | • Only a small sample of the entire docked bike users. • Many objective features |
| Chung et al., 2024 (Seoul, South Korea) | 74,397 bikeshare trips; one day, Oct.21st, 2021 | Shortest path | (−)Distance;(−) Bike Boulevards (+) Buffered Bike Lane;(−) Shared Sidewalks Bikeways;(−) Vehicle Lane | **N/A** | (−) Floating Population | Detour Percentage. PSL (*R2* 0.854) | • Limited time |
| Dockless Bikeshare Trajectories | | | | | | | |
| Zhao et al., 2024 (Xiamen, China) | 53,460 trajectories; 6-10am, 2020.11.22 | Shortest path | (+) Density of POIs; (+)street type (+) Density of bicycle parking POI; street network; etc. | (+) streetlights; (+)traffic lights; (insignificant) Street greenery | **N/A** | Preferred Street Index. GWR (*R2* 0.145—0.325) | • a single day observation • Lack of perceptual variables. |
| This Study(Ithaca, USA) | >4,400 trajectories; 2019.11—2020.03 | Shortest path | | Objective features; Subjective Perceptions | population density, etc. | Desirable Bikeshare Index. RF | |

**Table A2**
Descriptive summary of all variables (n = 2127).

| Var. name | Description | Source | Mean | S.D. | Max | Min |
|---|---|---|---|---|---|---|
| ***Dependent variable*** | | | | | | |
| DBI | Desirable Bikeshare Index | | 67.033 | 98.069 | 766.000 | 2.000 |
| ***Independent variable*** | | | | | | |
| ***Macro-level Built Environment (BE)*** | | | | | | |
| BTA800c | Betweenness Angular 800c | QGIS − sDNA | 1100550.919 | 2938025.982 | 24448650.000 | 0 |
| MAD800c | Mean Angular Distance 800c | | 757.712 | 405.414 | 2702.217 | 0 |
| LLen | Line length | | 0.000 | 0.000 | 0.003 | 0 |
| LBear | Line Bearing | | 155.207 | 128.183 | 359.950 | 0 |
| Commercial | Commercial Point of interest | Open Street Map. Count in 50 m buffer along the road segment. | 1.836 | 4.032 | 29.000 | 0 |
| Community | Community Point of interest | | 0.266 | 0.571 | 4.000 | 0 |
| Forest | Forest Point of interest | | 0.003 | 0.061 | 2.000 | 0 |
| Industrial | Industrial Point of interest | | 0.004 | 0.075 | 2.000 | 0 |
| Public Service | Public Service Point of interest | | 0.013 | 0.114 | 1.000 | 0 |
| Recreation | Recreation Point of interest | | 0.034 | 0.205 | 3.000 | 0 |
| Residential | Residential Point of interest | | 3.291 | 6.341 | 34.000 | 0 |
| R_C_ratio | Residential-to-Commerial Ratio | | 1.316 | 3.388 | 27.333 | 0 |
| POI Count | Point of interest | | 0.221 | 0.557 | 5.000 | 0 |
| MEAN | Mean road slope calculated by GIS | DEM from USGS | 3.725 | 5.043 | 59.966 | 0.252 |
| Bike lanes dum | Bike lanes dummy variable | Open Street Map | 0.875 | 0.330 | 1.000 | 0 |
| Primary dum | Primary road dummy variable | | 0.083 | 0.276 | 1.000 | 0 |
| Secondary dum | Secondary road dummy variable | | 0.031 | 0.172 | 1.000 | 0 |
| ***Sociodemographic*** | | | | | | |
| Vehicle_po | Vehicle Population | American Community Survey (ACS)-Aggregated from each census block to road segment. | 0.309 | 0.295 | 1.035 | 0 |
| Median Inc | Median Income | | 33060.341 | 33396.122 | 99333.000 | 0 |
| PopDen | Population Density | | 3079.633 | 5252.047 | 27896.437 | 0 |
| WorkDen | Employee Density | | 3009.296 | 16887.292 | 176883.431 | 0 |
| TotHH | Total Households | | 61.097 | 117.033 | 919.000 | 0 |
| ***Micro-level Street Environment (SE)*** | | | | | | |
| *Objective feature* | | | | | | |
| % Road damage (Percentage) | Crack area percentage detected by CNN | Google SVIs in 600 x 400 pixels | 19.586 | 11.685 | 75.000 | 0 |
| Tree | % of pixels corresponding to a specific visual element classified by PSPNet. Vehicle count using MaskRCNN | | 0.236 | 0.179 | 0.797 | 0 |
| Car | | | 0.013 | 0.029 | 0.241 | 0 |
| Earth | | | 0.011 | 0.041 | 0.440 | 0 |
| Grass | | | 0.091 | 0.104 | 0.587 | 0 |
| Person | | | 0.000 | 0.002 | 0.018 | 0 |
| Ashcan | | | 0.000 | 0.001 | 0.019 | 0 |
| Awning | | | 0.000 | 0.002 | 0.041 | 0 |
| Building | | | 0.080 | 0.108 | 0.971 | 0 |
| Sidewalk | | | 0.029 | 0.048 | 0.379 | 0 |
| Streetlight | | | 0.001 | 0.002 | 0.035 | 0 |
| Signboard | | | 0.001 | 0.004 | 0.050 | 0 |
| Water | | | 0.001 | 0.010 | 0.304 | 0 |
| ct_traffic | | | 0.219 | 0.824 | 18.000 | 0 |
| *Subjective perception* | | | | | | |
| Order | | Visual surveys and Google SVIs in 600 x 400 pixels. | 0.502 | 0.052 | 0.682 | 0.325 |
| Accessibility | Expert feedback on visual surveys and ML predicts. | | 0.514 | 0.097 | 0.816 | 0.238 |
| Aesthetic | | | 0.541 | 0.075 | 0.703 | 0.289 |
| Ecology | | | 0.569 | 0.110 | 0.866 | 0.145 |
| Enclosure | | | 0.507 | 0.145 | 0.797 | 0.071 |
| Richness | | | 0.472 | 0.086 | 0.705 | 0.247 |
| Scale | | | 0.504 | 0.080 | 0.723 | 0.248 |

**Table A3**
OLS Regression Results.

| Variable | Coefficient | Std Err | t | P>\|z\| |
|---|---|---|---|---|
| Intercept | 99.6819 | 20.951 | 4.758 | 0 |
| ***Macro-level SE*** | | | | |
| BTA800c | 0.000005143 | 0.000000642 | 8.017 | 0 |
| MAD800c | −0.0231 | 0.009 | −2.63 | 0.009 |
| LLen | −10940 | 6191.685 | −1.767 | 0.077 |
| LBear | −0.0042 | 0.018 | −0.236 | 0.814 |
| Commercial | 5.3284 | 0.676 | 7.884 | 0 |
| Community | 7.9137 | 3.281 | 2.412 | 0.016 |
| Forest | 13.5628 | 28.972 | 0.468 | 0.64 |
| Industrial | −5.0397 | 23.789 | −0.212 | 0.832 |
| Public Service | −1.8119 | 15.829 | −0.114 | 0.909 |
| Recreation | 35.0295 | 9.175 | 3.818 | 0 |
| Residential | 1.0938 | 0.461 | 2.374 | 0.018 |
| R_C_ratio | 0.5305 | 0.75 | 0.707 | 0.479 |
| POI Count | 35.7778 | 3.702 | 9.664 | 0 |
| MEAN | 1.332 | 0.397 | 3.357 | 0.001 |
| Bike lanes dum | −1.107 | 7.439 | −0.149 | 0.882 |
| Primary dum | −6.0121 | 7.019 | −0.857 | 0.392 |
| Secondary dum | 10.8419 | 10.811 | 1.003 | 0.316 |
| ***Sociodemographic*** | | | | |
| Vehicle_po | 6.6699 | 9.859 | 0.677 | 0.499 |
| Median Inc | 0.0000001989 | 0.0000862 | −0.002 | 0.998 |
| PopDen | 0.0008 | 0.00 | 1.653 | 0.099 |
| WorkDen | 0.0002 | 0.00 | 1.516 | 0.13 |
| TotHH | −0.0664 | 0.023 | −2.919 | 0.004 |
| ***Micro-level SE (subjective perceptions)*** | | | | |
| Order | 63.9651 | 24.86 | 2.573 | 0.01 |
| Accessibility | 48.7349 | 18.318 | 2.66 | 0.008 |
| Aesthetic | −60.7863 | 25.827 | −2.354 | 0.019 |
| Ecology | −69.8192 | 18.765 | −3.721 | 0 |
| Enclosure | −3.2349 | 10.509 | −0.308 | 0.758 |
| Richness | 0.3494 | 20.501 | 0.017 | 0.986 |
| Scale | 59.0775 | 27.749 | 2.129 | 0.033 |
| ***Micro-level SE (objective streetscape features)*** | | | | |
| % Road damage | −0.2511 | 0.165 | −1.526 | 0.127 |
| Tree | 36.257 | 16.235 | 2.233 | 0.026 |
| Car | −6.5986 | 68.162 | −0.097 | 0.923 |
| Earth | −62.7472 | 48.591 | −1.291 | 0.197 |
| Grass | −28.2627 | 24.821 | −1.139 | 0.255 |
| Person | 2681.507 | 1061.137 | 2.527 | 0.012 |
| Ashcan | −130.523 | 2473.992 | −0.053 | 0.958 |
| Awning | 460.5512 | 911.42 | 0.505 | 0.613 |
| Building | −186.552 | 31.863 | −5.855 | 0 |
| Sidewalk | 20.9248 | 45.8 | 0.457 | 0.648 |
| Streetlight | 1791.729 | 856.407 | 2.092 | 0.037 |
| Signboard | −583.938 | 487.758 | −1.197 | 0.231 |
| Water | 229.0617 | 181.674 | 1.261 | 0.208 |
| ct_traffic | 13.7083 | 2.305 | 5.946 | 0 |
| ***R2.*** | *0.31* | | | |


**Fig. A1.** Bee-swarm Plot of Features from SHAP.

## Appendix B

### *Actual routes data processing*

Actual trajectory data was provided by Lime dockless bikeshare service as part of an exclusive partnership with the City of Ithaca initiated in autumn 2017. Lime launched 350 dockless bicycles in 2018, tracked via their mobile app with integrated GPS, with each trip assigned a unique identifier, geospatial coordinates, and timestamps. The dataset covers trips from November 21, 2019, to March 17, 2020, initially comprising 5,038 trips, refined to 4,430 trips to reflect commuter behavior based on secondary data from Qiu and

Chang (2021). The data processing was conducted in QGIS. After importing the raw GPS data into QGIS, we observed a certain deviation between the GPS coordinates and the road network data downloaded from OSM, which was attributed to the urban canyon effect (Chen et al., 2020). To correct this deviation, we applied a map-matching algorithm to realign the GPS data with the road network. Perpendicular lines were drawn from the GPS coordinates to the road network, and each trajectory coordinate was then aligned with the nearest point on the road network by finding the shortest perpendicular distance. These points were then collected and reconnected to form an accurate trajectory network.

$$P_t' = argmin(verticaldis(P_t, R_i)\,), i/in\{1, 2, ..., 32649\}$$

where, $P_t$ is the raw GPS point; $P_t'$ t is the matched point on the road grid; $R_i$ is the road segmentation identity; *verticaldis* is the function that computes the vertical distance between the raw GPS point and road segmentation.

Dijkstra’s algorithm was further employed for route reconstruction to determine the shortest paths within the road network, aligning closely with actual cycling routes and contributing to a detailed analysis of cycling volumes on individual road segments. The GPS data's high precision ensured that the algorithm's shortest paths closely approximated the cyclists' actual routes. This methodology aligns with its prevalent use in urban studies to identify the most efficient routes within a graph's structure (Risald & Suyoto, 2017; Salazar Miranda et al., 2021). Notably, the algorithm's implementation had a minimal influence on cyclists' selection of specific roads with distinctive attributes.

```
Function Dijkstra (graph, source):
distance:= map with default value of ∞
distance[source]:= 0
Q:= priority queue containing all nodes in graph
visited:= empty set
while Q is not empty:
u:= Q.extract_min()
if u in visited:
continue
visited.add(u)
for each neighbor v of u:
alt:= distance[u] + weight(u, v)
if alt < distance[v]:
distance[v]:= alt
Q.decrease_key(v, alt)
return distance
```

*where Dijkstra's algorithm initializes the distance to the source node as zero and all other nodes as infinity, while also initializing a priority queue containing all nodes. The algorithm repeatedly extracts the node with the smallest distance from the priority queue, updates the distances to its neighboring nodes if a shorter path is found, and marks the node as visited. This process continues until the priority queue is empty. The result is a dictionary that maps each node to its shortest path distance from the source.*

## Appendix C

*Micro-level SE: objective streetscape features*

Google Street View has emerged as a prevalent data source for capturing micro-level environment attributes at the street scale, with its reliability supported in transportation and environmental studies (Vanwolleghem et al., 2014; Song et al., 2024). Utilizing the Google Street View Static API, images were systematically collected from specific coordinates, ensuring consistency with standardized camera settings of a 120-degree field of view and a resolution of 600 × 400 pixels (Ito and Biljecki, 2021; Qiu et al., 2022). A dataset of 11,426 valid street view images (SVIs) was compiled after removing unsuitable views (Qiu et al., 2022; Song et al., 2023).

For feature extraction, the Pyramid Scene Parsing Network (PSPNet), pre-trained on the ADE20K dataset, was utilized for its superior pixel-level accuracy in urban scenes. Its extensive collection of urban scenes from over 50 cities makes it exceptionally suitable for the semantic segmentation of urban landscapes (Zhou et al., 2018). This model identified 31 distinct visual categories within the images, enhancing the granularity of our urban environmental analysis (Zhao et al., 2017). The dataset was refined to include 12 critical visual elements, such as buildings, sidewalks, and vegetation, for further analysis. The View Index quantified the prevalence of these features by calculating the ratio of pixels representing each element to the total pixels in each SVI (Zhang et al., 2018).

## Appendix D

*Micro-level SE: subjective perceived qualities*

Over 300 SVIs were randomly selected for evaluation by 50 design experts, aged between 20 and 35, with an approximately 1:1

gender ratio. All survey participants had formal training in design disciplines, so they are well-versed in the terminology of the perceived qualities (Qiu et al., 2022). In this study, each image received an average of 22.5 pairwise comparison votes, exceeding the conventional threshold (i.e., 12 to 36) recommended in the literature (Salesses et al., 2013) and the suggested 22 pairwise comparison criteria to ensure robustness (Gu et al., 2025). Through pairwise comparison surveys, survey participants were asked to make judgments on each SVI across seven perceived qualities to choose the one of better quality: Order, Accessibility, Aesthetics, Ecology, Enclosure, Richness, and Scale. Scores from experts were converted and standardized using the Microsoft TrueSkill algorithm into a 0 to 1 scale. For the predictive modeling of perceived qualities, the dataset was expanded to encompass over 300 SVIs and their ranked scores, serving as dependent variables in predictive models. Objective view indices from the SVIs acted as independent variables. The dataset was divided into training (80%) and testing (20%) subsets. Multiple machine learning algorithms were tested, including Random Forest (RF) and K-Nearest Neighbors (KNN), to predict the seven design qualities. The algorithms' performance was evaluated using metrics like R2, MAE, RMSE, and MAPE. The RF model emerged as the most effective and was subsequently adopted to project subjective perceptions across the full dataset. The trained models achieved R2 of up to 0.6, which is comparable to the performance of prior similar studies (Chen & Biljecki, 2023; Qiu et al., 2022; Song et al., 2024), and the details are discussed in a prior project (Tian et al., 2021). Nevertheless, we acknowledge that a method of limitation regarding formal inter- and intra-rater reliability assessments was not conducted in this study. However, our approach aligns with established protocols in large-scale perception studies (Dubey et al., 2016). According to the TrueSkill algorithm's documented performance characteristics, rankings derived from 22 to 32 votes per image achieve transitivity (internal consistency) exceeding 75% which suggests this technical mechanism can partially offset the absence of formal reliability checks (Salesses et al., 2013). Nevertheless, future research should incorporate systematic reliability testing procedures to further enhance the validity and reproducibility of the evaluation results (Gu et al., 2025).

## Appendix E

*Crack Detection using CNN*

The CNNs, which have demonstrated high accuracy and precision in image classification and recognition tasks, present a promising framework for addressing this issue (Eisenbach et al., 2017). A open-source imagery dataset of labelled road cracks (Özgenel, 2019) was obtained for road damage detection, consisting of 20,000 positive samples and 20,000 negative samples. Given the limited number of training samples, we employed a pre-trained network and applied image augmentation techniques (e.g., flipping, rotation, and brightness variation) to enhance detection performance. Image augmentation effectively addresses sample insufficiency by increasing sample size and aiding model generalization (Maeda et al., 2021).

Each RGB image (227 × 227 pixels) in the training dataset was categorized into positive (cracked) and negative (no cracked) samples, further being split into training and validation subsets. Random transformations were applied to introduce variability. We allocated 85% of the data to the training set and 15% to the validation set. The neural network architecture was based on a pre-trained ResNet-50 model from ImageNet (He et al., 2015), utilizing transfer learning to adapt the model for a binary classification task. The base layers of the ResNet-50 were frozen, while the final fully connected layer was modified to fit the task-specific output (Dwivedi, 2019). The model was trained on the training set, and its performance was monitored on the validation set using loss and accuracy metrics. Additionally, GPU acceleration was employed, and learning rate scheduling was applied to optimize training efficiency and ensure convergence.

## Appendix F

*KANO model – three-factor theory*

The Kano model is based on the asymmetric relationship between product/service attributes and customer satisfaction. The theory systematically classifies attributes into three categories (Kano et al., 1984):

Basic factor (Must-be Quality): These represent the minimum requirements that people take for granted, meaning that when absent or poorly executed, they lead to significant dissatisfaction. However, when exceeding a sufficient threshold, they do not proportionally increase satisfaction. It is characterized by steep dissatisfaction at low performance levels and a gradual transition toward neutral satisfaction when sufficiently provided, forming an asymmetric curve concentrated in the negative satisfaction region.

Excitement/Delight factor (Attractive Quality): These represent unexpected features that exceed customer expectations. When absent, they do not lead to dissatisfaction because customers do not expect them. When present, especially at high levels, they generate disproportionate satisfaction and delight. This is characterized by neutral satisfaction at low performance levels, and when sufficiently provided, it steeply accelerates people to high satisfaction, forming an asymmetric curve concentrated in the positive satisfaction region.

Performance factor (One-dimensional Quality): These represent attributes where customer satisfaction changes proportionally with performance level. Better performance leads to higher satisfaction, while worse performance results in lower satisfaction, characterized by a linear or near-linear correlation between performance level and satisfaction.

In our study, factor types are identified by comparing the shapes of PDP curves with these theoretical patterns (Fig. F1).

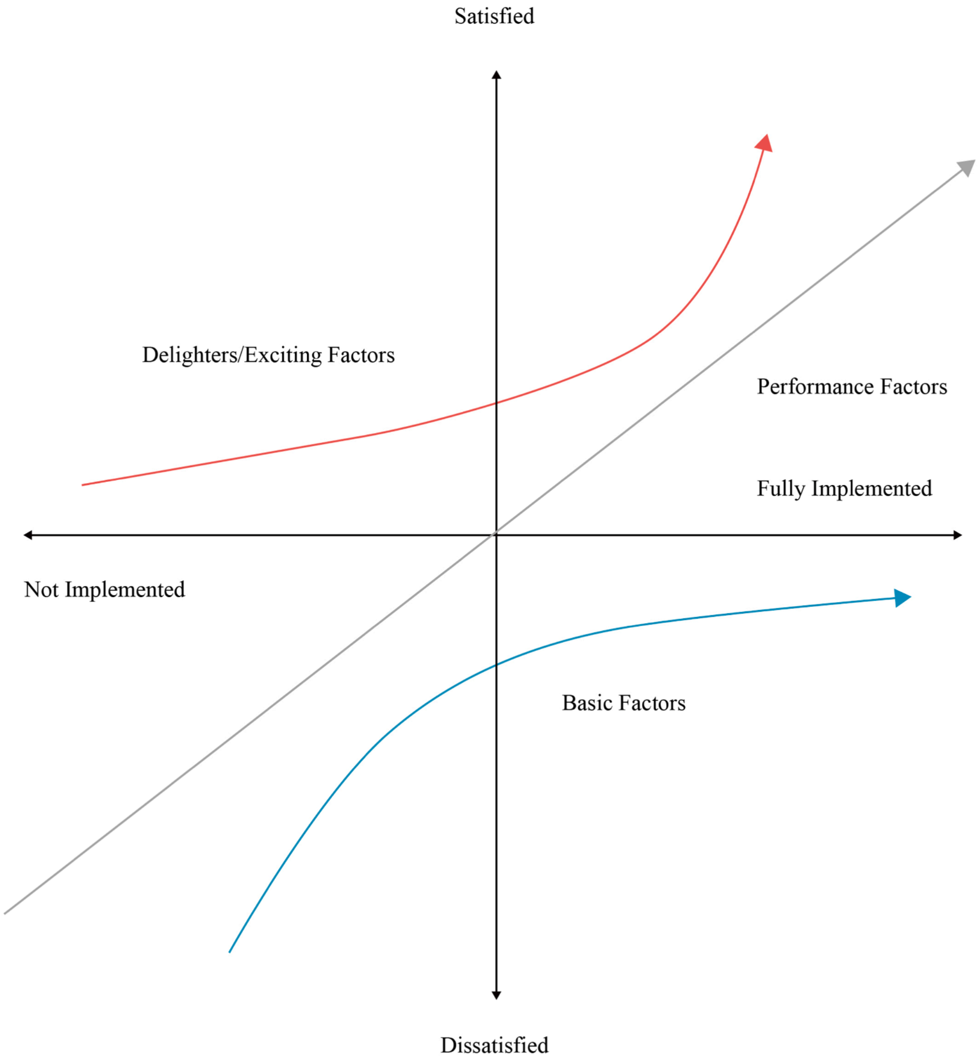



**Fig. F1.** KANO Model – Three-factor Theory Trend

## Appendix G. Streetscape cost estimates table

| Streetscape variable | Unit cost (Range) | Unit type | Source (city/agency) | Year | Estimated cost per mile |
|---|---|---|---|---|---|
| Street Trees (Planting + Soil + Establishment) | $300 – $4,350 per tree | each (installed; higher end includes 5-year care) | Long Beach Public Works (Manager, 2023); Los Angeles Board of Public Works (CityWatch Los Angeles, 2022) | 2023; 2022 | $600,000 – $870,000 per mile (assume 200 trees per mile, 8 m apart) |
| Sidewalk Installation (5′ Wide Concrete) | $25 – $85 per linear foot | per linear foot (one side) | Victoria Transport Policy Institute (VTPI) (Litman, 2023) | 2023 | $132,000 – $448,800 per mile (Length = 1 mile = 5,280 ft.) |
| Bike Lane (Painted) | $15,000 – $80,000 per mile | per mile (striping & signs on existing road) | VTPI (Litman, 2023) | 2023 | $15,000 – $80,000 per mile |
| Bike Lane (Protected/Separated) | $133,000 – $537,000 per mile | per mile (protected lane with barrier) | Pedestrian & Bicycle Info. Center (via FHWA) (Streetsblog, 2020) | 2020 | $133,000 – $537,000per mile |
| Road Resurfacing (Asphalt Overlay) | $110,000 – $140,000 per lane-mile | per lane-mile (2″ asphalt overlay) | Arkansas DOT (planning average) (Arkansas Department of Transportation, 2018) | 2018 | $220,000 – $280,000 per mile (for 2-lane roads) |

(continued on next page)

(*continued*)

| Streetscape variable | Unit cost (Range) | Unit type | Source (city/agency) | Year | Estimated cost per mile |
|---|---|---|---|---|---|
| Street Lighting (Led Lamp + Pole) | $2,500 – $3,500 per fixture | each (installed LED street light) | EcoSmart (Lighting industry guide) (EcoSmart., 2024) | 2024 | $87,500 – $122,500 per mile (assume 5,280 ft ÷ 150 ft ≈ 35 fixtures per mile) |
| Street Furniture – Bench | ~$2,080 each | per bench (installed) | City of Cambridge (PB Project) (City of Cambridge, 2017) | 2017 | $20,800 per mile (Approx. 10 benches per mile of active commercial/ pedestrian corridor) |

## Data availability

Data will be made available on request.

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